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Understanding Hard Disks & File Systems

INVESTIGATION OVERVIEW

Background. This examination covers three forensic disk images supplied as evidence in case CASE-2026-M3.

EV-001 comes from a **Linux workstation** recovered from a crime scene and suspected of holding material relevant to an investigation.

EV-002 and EV-003 come from **Windows machines** affected by an incident and were prepared by the organization's security team. The aims were to establish what each image contains, how it was configured and used, whether deleted or hidden data could be recovered, when events took place, and whether the recovered material is what it claims to be.

Approach. Each image was fingerprinted on receipt with two independent hashing methods and was only ever read, never written.

I then worked through four stages:

file-system structure and identity, recovery of deleted data (by metadata and by carving), timeline reconstruction, and signature verification of the recovered files. Every important result was reproduced with a second tool or method.



Principal findings.

1. **EV-001** is a single Ext4 volume from an **Ubuntu 16.04 LTS**4 virtual machine named ***roger-Virtual-Machine***, installed **on 9 June 2016** and configured for the **Asia/Kolkata time zone.** Its scheduled task file contains only default entries. The timestamps suggest the system was stopped without a clean shutdown **(Moderate confidence).**

2. **HideMarkedCols.xls**, the file of interest on **EV-001,** was introduced onto the system rather than created there. Its content-**modified date of 28 September 2011** is almost **five years older** than the system itself. Its metadata, raw disk location, two independent carving tools, file slack and the timeline all agree on its identity **(High confidence).**

   Note: The slack space contained only null bytes, so no residual data was found.

3. **EV-001 is an incomplete image.** It holds about **1.01 million of the 1.29 million blocks** the file system declares. The home directories of other users **(harry, jason, martha)** are named but cannot be read.

   Note: This is a limit of the evidence, not proof of deletion.

4. **EV-002** is a complete **2 GiB NTFS volume**. **22 deleted files were recovered** with names, paths and timestamps intact, and four methods agree on that count.

5. **Signature verification** of the **22 files** showed **14 matching** their extensions and **5 plain-text files with no signature to test.**

   Three Office documents (Word_Doc.docx, Word_Doc1.docx, Tutorial.pptx) **do not match:** they begin with a legacy Office header but **carry encrypted-package markers.** They were not opened or decrypted, and no conclusion about their content is drawn.

6. **A systematic timestamp irregularity** affects all **22 recovered entries:**

   The MFT-modified time is earlier than the creation time, and earlier than the volume's own creation. The cause is undetermined and is flagged for further study.

7. **EV-003** is an MBR disk with two NTFS partitions **(500 MiB at sector 2,048 and about 24.5 GiB at sector 1,026,048).** Its partition layout and file-system parameters were established, but its contents were not examined.

Limitations. The main limits are the **truncated EV-001 image**, an unexplained difference between WinHex's reported and actual carved-file counts **(58 against 232),** and the **unknown time zone of the Windows systems**. An early carving attempt that used the wrong source was discarded and excluded from all findings.

Integrity. All three images matched their original SHA-256 baselines at every re-check, including the final verification. The evidence is therefore shown to be unaltered.

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Scope, Objectives and Authority

The scenario supplied with the lab material describes a **dead system** recovered from a crime scene and suspected of holding material relevant to an investigation **(EV-001),** together with forensic images of Windows machines affected by an incident, prepared by the organization's security team **(EV-002 and EV-003).**

The examination **had five aims**, one per lab.

Lab 1 established the structure, identity and condition of the Linux file system and examined two files of interest.

Lab 2 did the same for the Windows images, examined the NTFS system files and recovered deleted files by the metadata method.

Lab 3 recovered data from the Linux image without relying on the file system, by carving, and examined the slack space of the file of interest.

Lab 4 built timelines from file-system metadata and used them to cross-check earlier results.

Lab 5 tested whether the recovered files were genuinely what their extensions claimed.

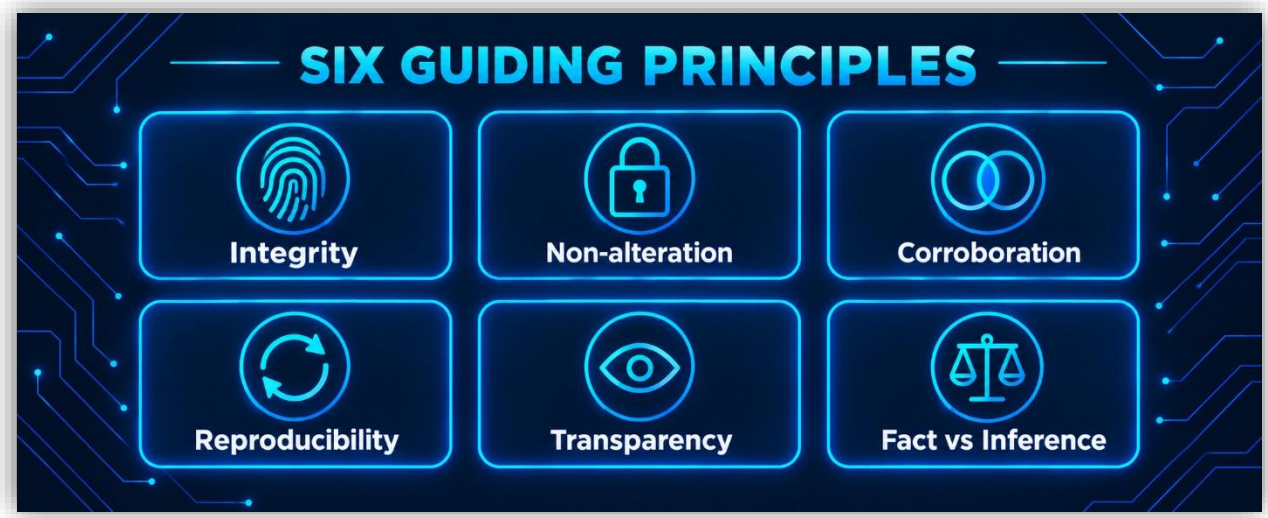
The examination deliberately did not cover registry, event-log or user-account analysis, e-mail or archive contents, opening or decrypting any recovered document, or timestamp-tampering analysis. Where such material was noticed, it is recorded in the Leads Register (Section 13) by name or count only.

Would be discussed later in upcoming modules

Methodology, Standards and Tools

The work followed **ISO/IEC 27037** for **identification, collection and preservation** of digital evidence, and **NIST SP 800-86** for integrating forensic techniques into an investigation. Six principles guided every step:

- **Integrity**: hashes were calculated at receipt and re-checked afterwards.
- **Non-alteration**: images were only ever read, never written. WinHex was set to read-only mode, PhotoRec showed its source as read-only, and all output went to a separate Exports folder.
- **Corroboration**: each important result was reproduced with a second tool or method.
- **Reproducibility**: the commands are listed in Appendix A.
- **Transparency**: errors, discarded runs and unresolved discrepancies are reported openly.
- **Fact and inference**: every finding states what was observed and then, separately, what it may mean. Interpretations carry a confidence level: High (directly observed and corroborated), Moderate (observed, with a reasonable interpretation) or Low (inference only).



Note: No separate working copy of any image was made. The images were examined in place using read-only access, and integrity was demonstrated by re-hashing rather than by working on a duplicate.

Tool	Used for
Autopsy 4.23.1	Case management, file browsing, ingest modules, deleted-file view
The Sleuth Kit	mmls, fsstat, img_stat, ifind, istat, icat, ffind, blkstat, fls, ils, tsk_recover
mactime (Perl)	Timeline generation from body files
WinHex (evaluation version)	Read-only raw inspection and header-based carving
PhotoRec 7.3-**WIP (Sept** 2025)	Second, independent carving tool
Windows PowerShell	Hashing, counting, inventory, signature and byte checks
certutil	Second hashing method
find, **findstr**, **Notepad**	Line counts and text searches

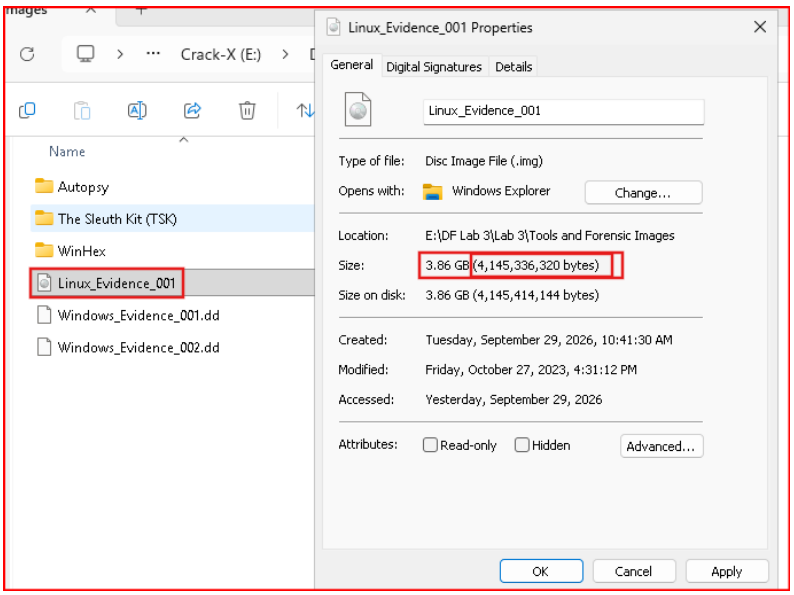
Securing Evidence: Identification and Integrity Baseline

A forensic examination is only as credible as the proof that the evidence was not changed, so the first task was to record what each image was and fingerprint it.

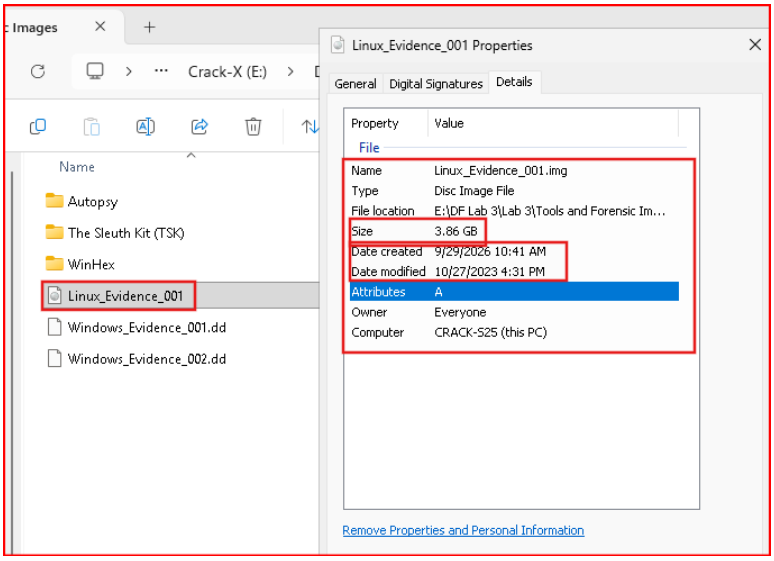
For EV-001, the file properties were recorded in Windows Explorer. The image is exactly **4,145,336,320 bytes**, which is **8,096,360 sectors of 512 bytes**. The properties also show that the file was copied to the examiner drive on **29 September 2026 at 10:41:30 AM**, while its own **modification date is 27 October 2023**. That the "Created" date is later than the "Modified" date is normal: copying a file gives the copy a new creation time while keeping the original modification time. The "Accessed" date changes whenever a tool reads the file, so it is not a sign of tampering either.



File properties, General tab (exact size in bytes)

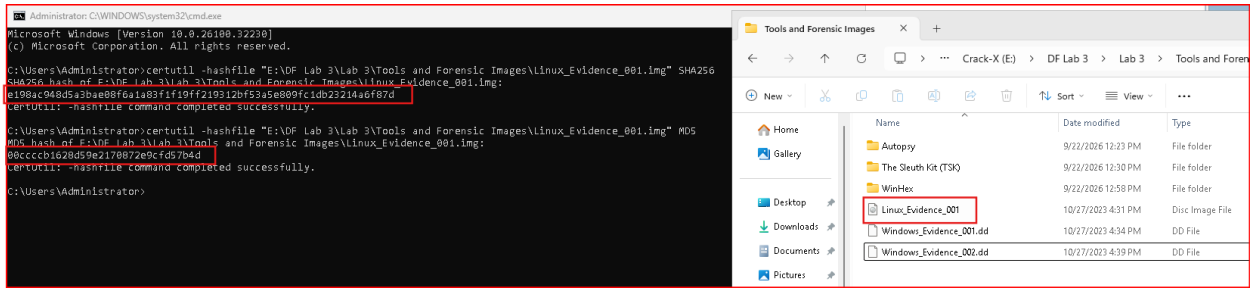


File properties, Details tab

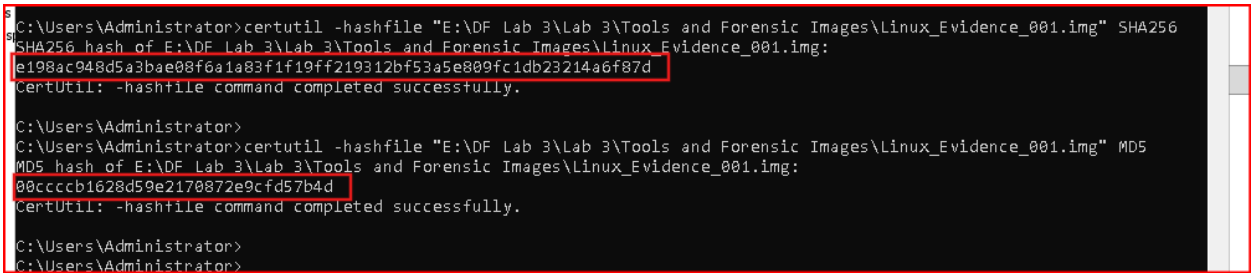


Both MD5 and SHA-256 were then calculated twice with certutil. Both runs produced identical values, which gave the baseline for every later check.

certutil output, run 1

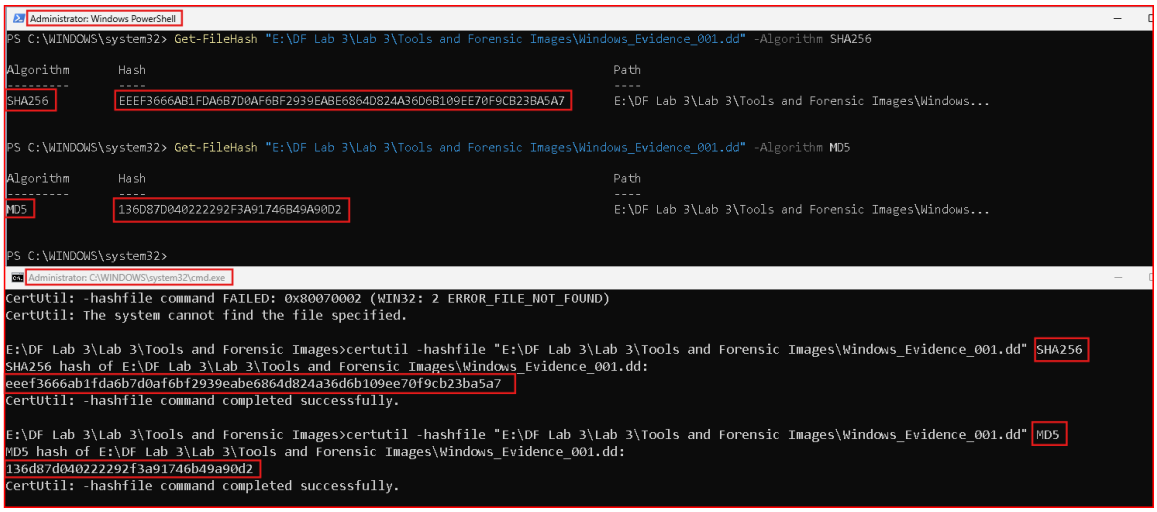


certutil output, run 2

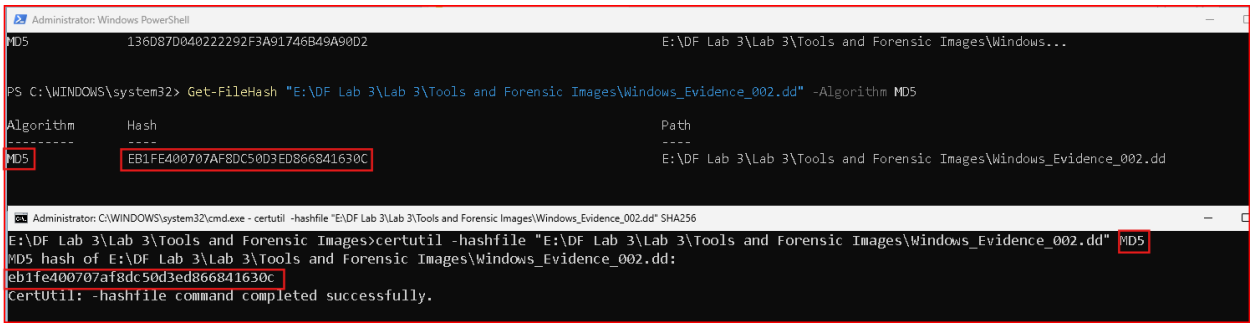


For the two Windows images, a different pair of tools was used. Each image was hashed with PowerShell's Get-FileHash and again with certutil. Matching results from two unrelated programs show more than repeatability, because the two programs are independent implementations. PowerShell prints hashes in upper case and certutil in lower case, so case was ignored when comparing.

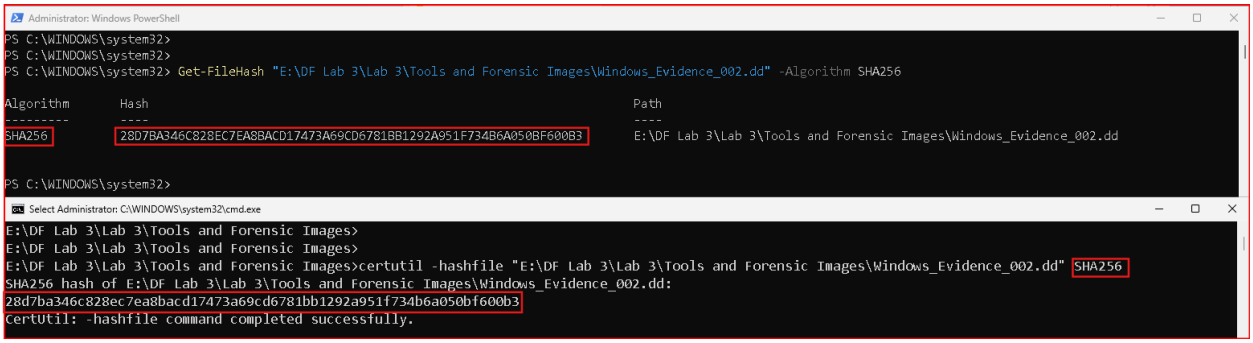
PowerShell and certutil, SHA-256 and MD5, Windows_Evidence_001.dd



PowerShell and certutil, MD5, Windows_Evidence_002.dd



PowerShell and certutil, SHA-256, Windows_Evidence_002.dd



	EV-001	EV-002	EV-003
File	Linux_Evidence_001.img	Windows_Evidence_001.dd	Windows_Evidence_002.dd
Layout	Bare Ext4	Bare NTFS	MBR, two NTFS partitions
Size (bytes)	4,145,336,320	2,147,483,648	26,843,545,600 (derived from mmls)
Sectors	8,096,360	4,194,304	52,428,800
MD5	00ccccb1628d59e2170872e9cfd57b4d	136d87d040222292f3a91746b49a90d2	eb1fe400707af8dc50d3ed866841630c
SHA-256	e198ac94…14a6f87d	eeef3666…b23ba5a7	28d7ba34…a050bf600b3

Integrity was re-verified at the end of completion. The final stage re-hashed all three images with SHA-256. Each result equaled the baseline.

Closing hashes:

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-FileHash "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          E198AC948D5A3BAE08F6A1A83F1F19FF219312BF53A5E809FC1DB23214A6F87D        E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img

PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-FileHash "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Windows_Evidence_001.dd" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          EEEF3666AB1FDA6B7D0AF6BF2939EABE6864D824A36D6B109EE70F9CB23BA5A7        E:\DF Lab 3\Lab 3\Tools and Forensic Images\Windows_Evidence_001.dd

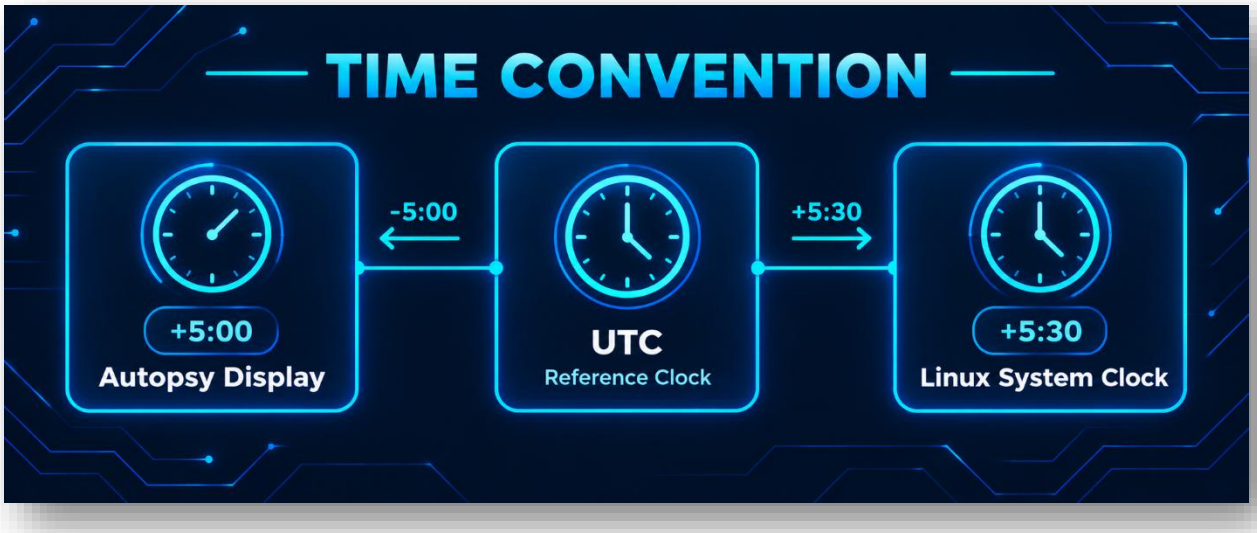
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-FileHash "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Windows_Evidence_002.dd" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          28D7BA346C828EC7EA8BACD17473A69CD6781BB1292A951F734B6A050BF600B3        E:\DF Lab 3\Lab 3\Tools and Forensic Images\Windows_Evidence_002.dd
```

Time convention

Several clocks appear in this work, so a single convention was fixed. All image-internal times are given in UTC for comparison with the display time alongside. Autopsy was configured to show **Asia/Tashkent time (UZT, UTC+5:00).**

The Linux system itself was **set to Asia/Kolkata (IST, UTC+5:30),** so its own clock ran 30 minutes ahead of what Autopsy displayed. The workstation **runs West Asia Standard Time, which equals UZT numerically.** The time zone of the Windows systems is not known, because it is held in the registry, which this module did not examine. For them only UTC and display time are given. Times of the examiner's own actions are workstation local time.



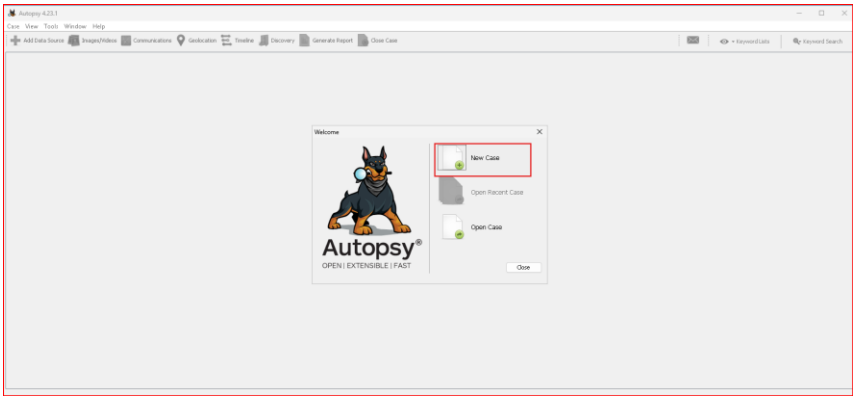
Understanding the Linux Image (EV-001)

Creating the case

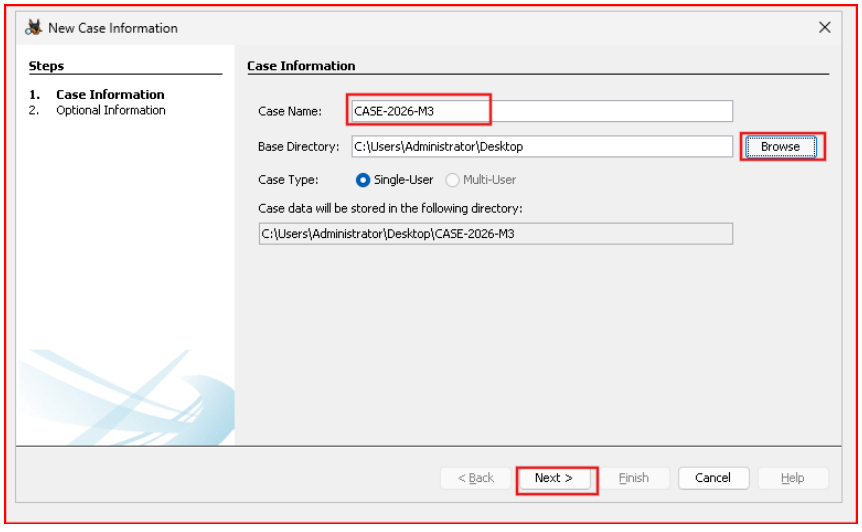
EV-001 was loaded into an Autopsy case named **CASE-2026-M3** (case number Case-002, single-user) with **Abdul Wahid** recorded as examiner. It was added as a disk image with the display time zone set to **Asia/Tashkent** and the sector size left on auto-detect. All default ingest modules were run except Keyword Search.

Autopsy recorded a device ID of **84a5404c-20ae-4252-8354-01552169d417** and a sector size of **512 bytes** and parsed the image under the host "Linux_Evidence_001.img Host". The directory tree that appeared contained the standard Linux root directories, including **bin, boot, etc, home, root, usr and var.**

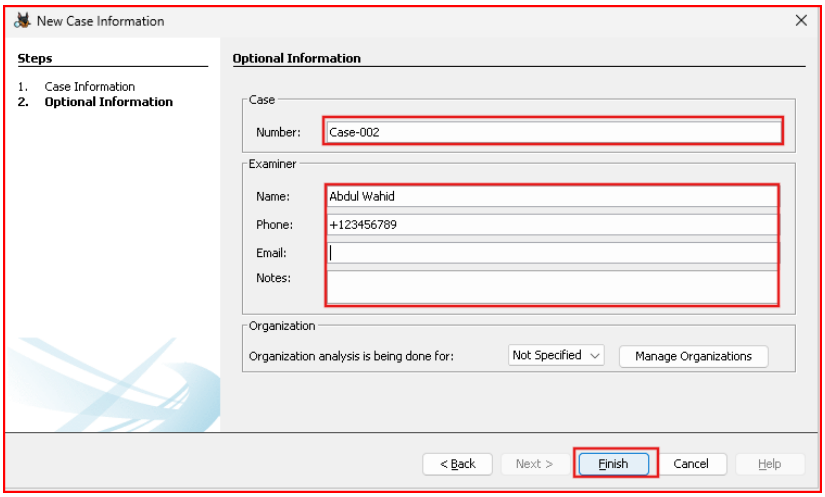
Autopsy welcome screen, New Case



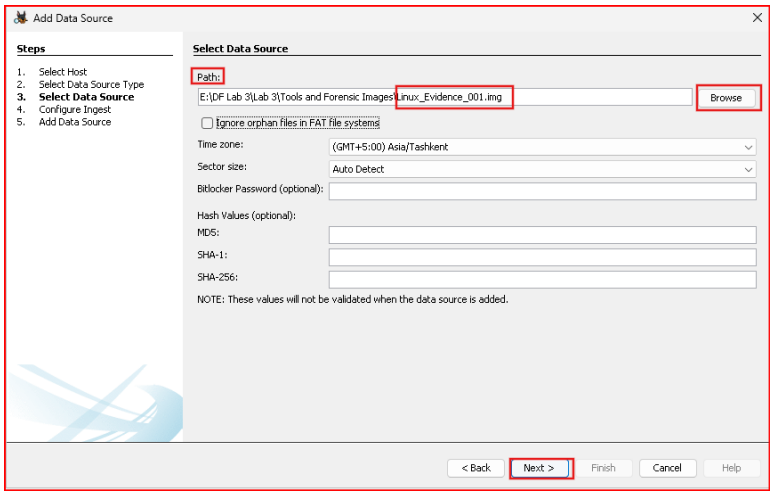
New Case Information, case name and base directory



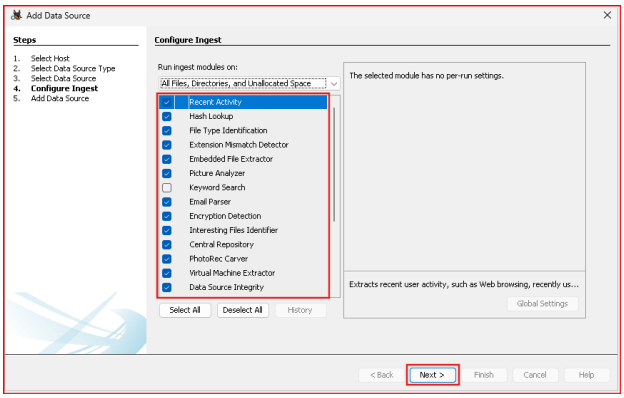
Optional information, case number and examiner



Add Data Source, image path and time zone



ingest module selection



directory tree after ingest

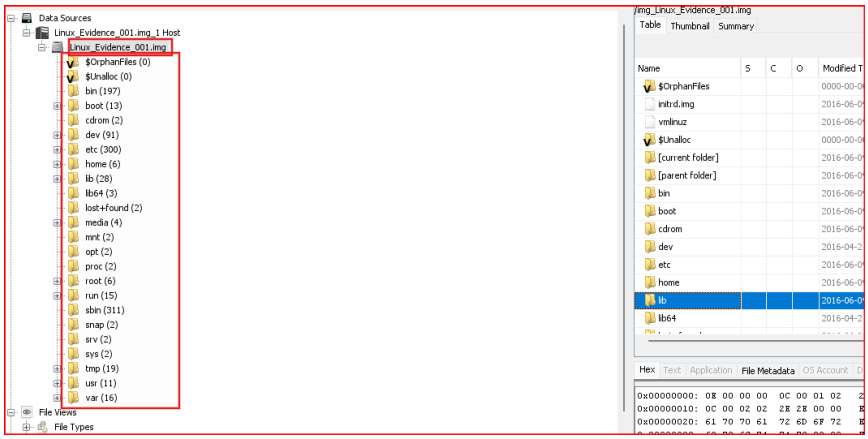
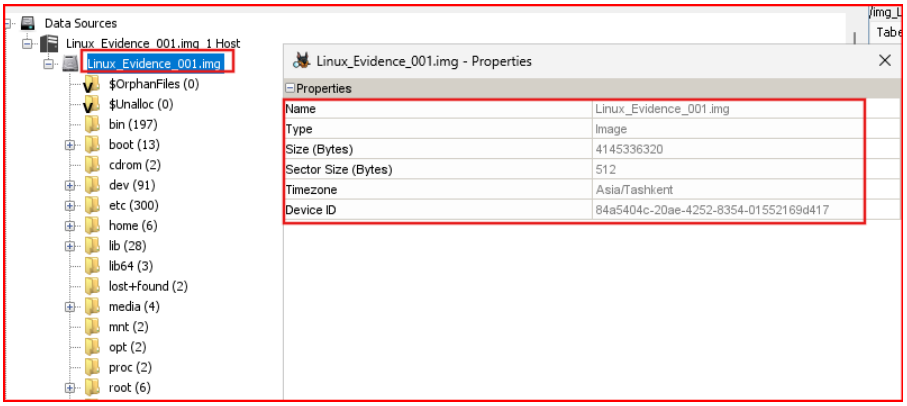


image properties window



Is there a partition table?

Before reading any file, the examiner needed to know how the image is laid out, because the layout decides whether commands need an offset. Running **mmls returned "Cannot determine partition type".** That means there is **no partition table** at all, and EV-001 is a single extracted volume rather than a whole disk.

As a result, no offset applies to any command run against it. This agrees with a comment inside the system's own **/etc/fstab,** which says root was on /dev/sda1 during installation. The original disk layout is not preserved in the image.

mmls output

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>mmls "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img"
Cannot determine partition type
```

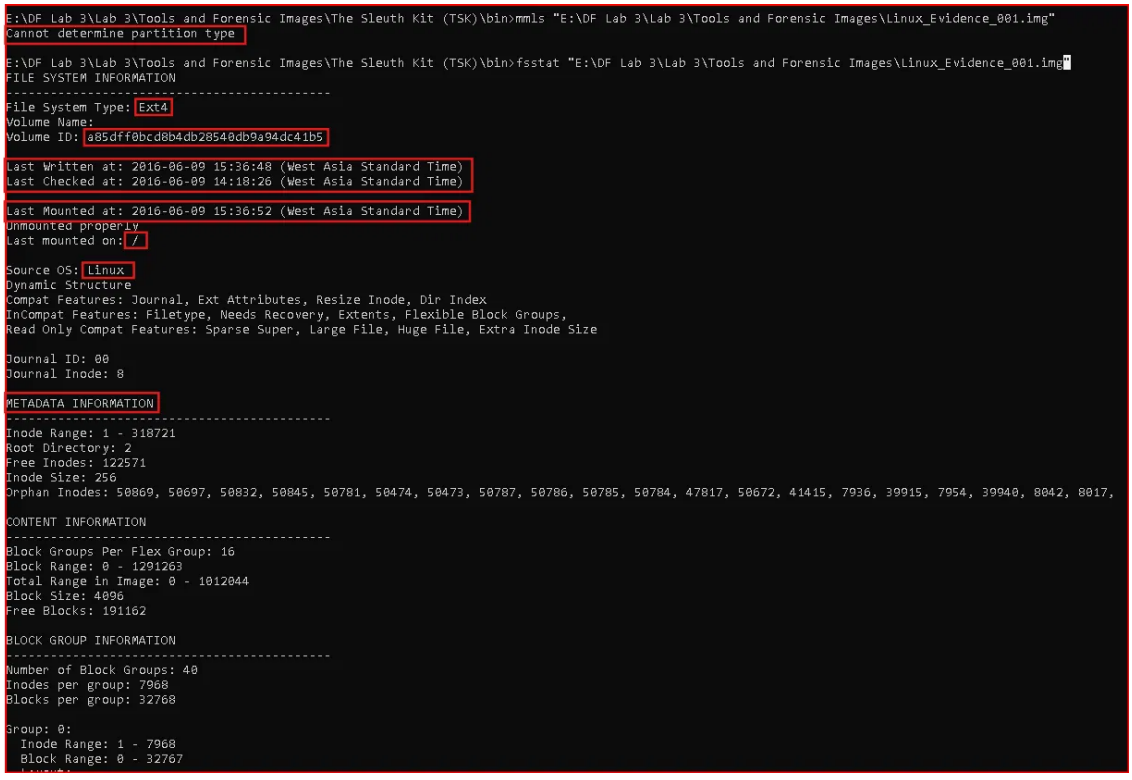
Identifying the file system

fsstat identified the file system as Ext4 and reported its parameters: a **4,096-byte block size, 256-byte inodes, inodes numbered 1 to 318,721, 40 block groups, a journal in inode 8,** and a **volume ID of a85dff0bcd8b4db28540db9a94dc41b5.**

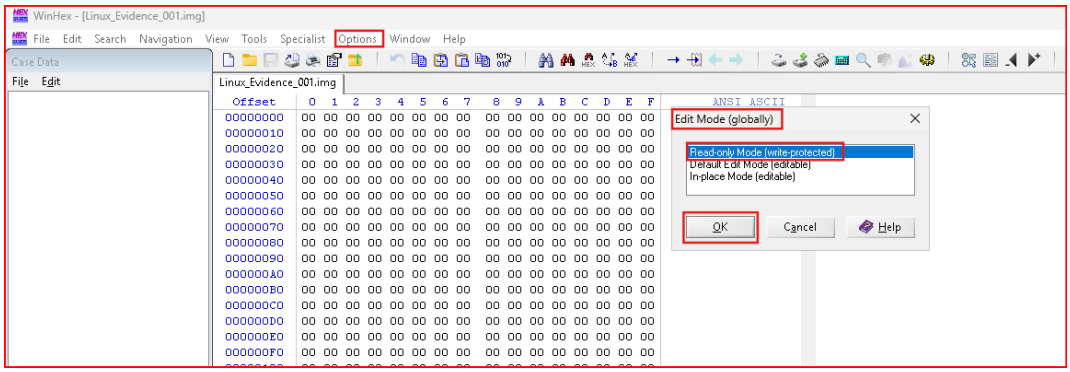
Per fsstat, the file system was last checked at **14:18:26**, last written at **15:36:48** and **last mounted at 15:36:52 (UZT) on 9 June 2016**.

To avoid trusting a single parser, the type was also confirmed from the raw bytes. **WinHex** was first placed in read-only (write-protected) mode. A Ext4 superblock begins **1,024 bytes** into the volume, and its magic number (0xEF53, stored little-endian as 53 EF) sits 56 bytes inside it, at offset 1,080 (0x438). The bytes **53 EF** were found at exactly that offset. The type was therefore confirmed independently of Autopsy and the Sleuth Kit.

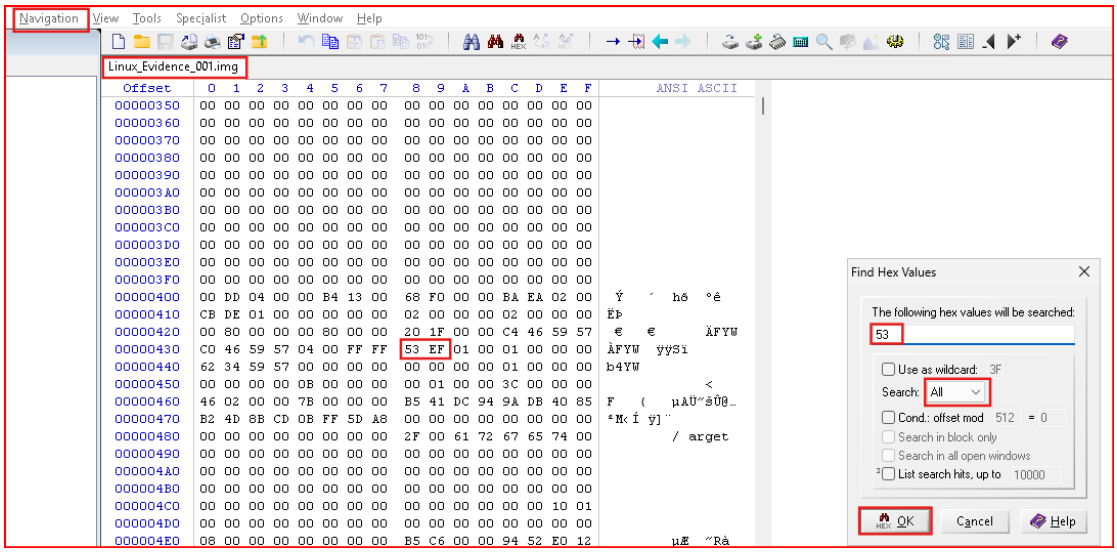
fsstat output



WinHex, Edit Mode set to Read-only (write-protected)



WinHex at offset 0x438 showing 53 EF



Who owned this machine and how was it configured?

The configuration files under /etc were read to establish identity.

lsb-release and issue show Ubuntu 16.04 LTS (xenial).

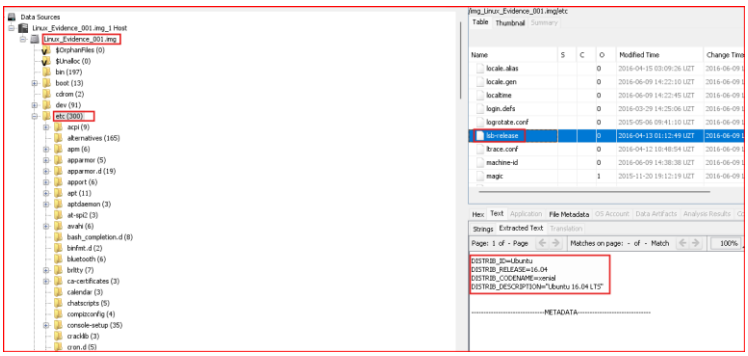
The **hostname** file contains **roger-Virtual-Machine**, and hosts hold only the standard localhost and IPv6 entries. The name points to a virtual machine used by an account called roger, which fits the roger home directory.

Note: Autopsy's Operating System Information view independently reports the same Ubuntu release.

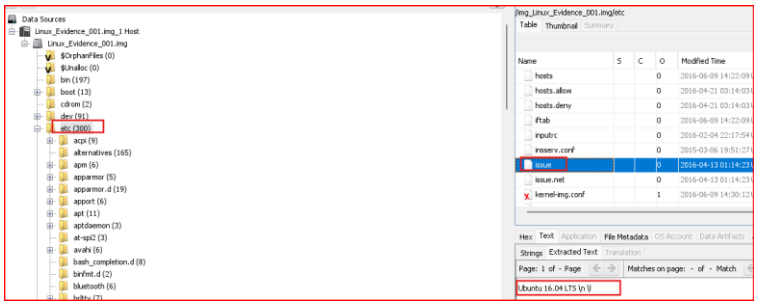
/etc/timezone contains Asia/Kolkata. /etc/fstab lists the root file system by UUID (b541dc94-9adb-4085-b24d-8bcd0bff5da8, ext4, remount read-only on errors) and a floppy-drive entry marked noauto. There is no swap, external or network mount configured, so no configured path to removable or network storage exists on this system.

Note: One unresolved oddity should be noted. The UUID in fstab differs from the volume ID that fsstat reports (a85dff0b…). This could result from a cloned or recreated file system or from how the image was prepared, but the evidence does not decide between them, so no conclusion is drawn.

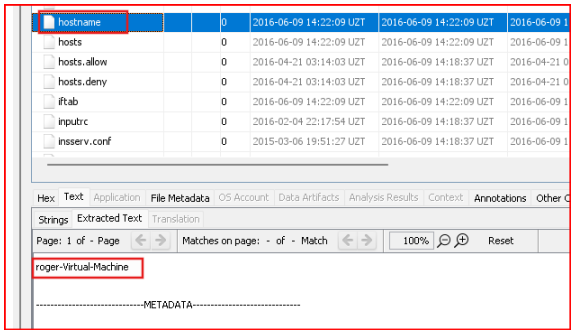
lsb-release content



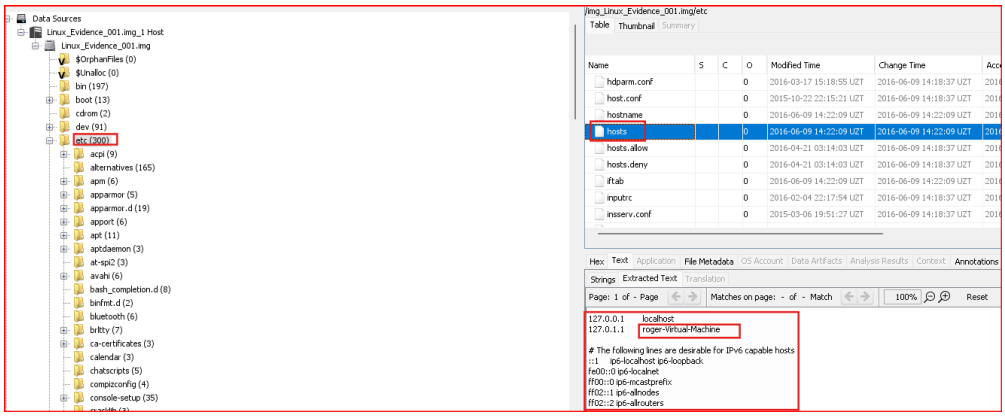
issue content



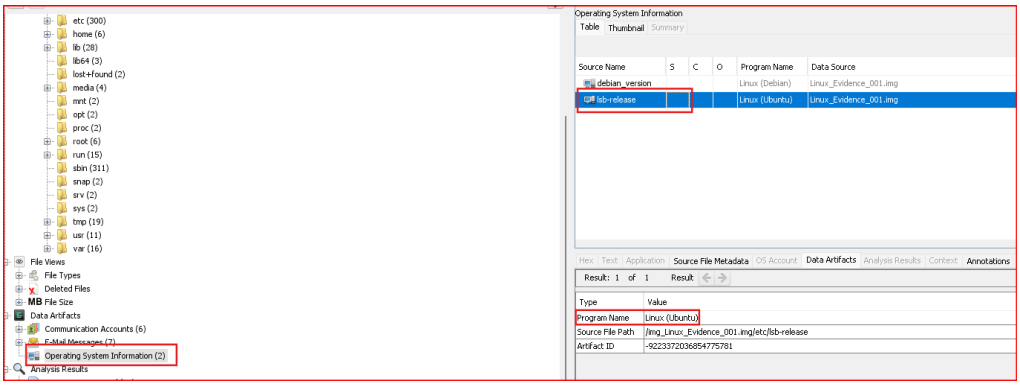
hostname content



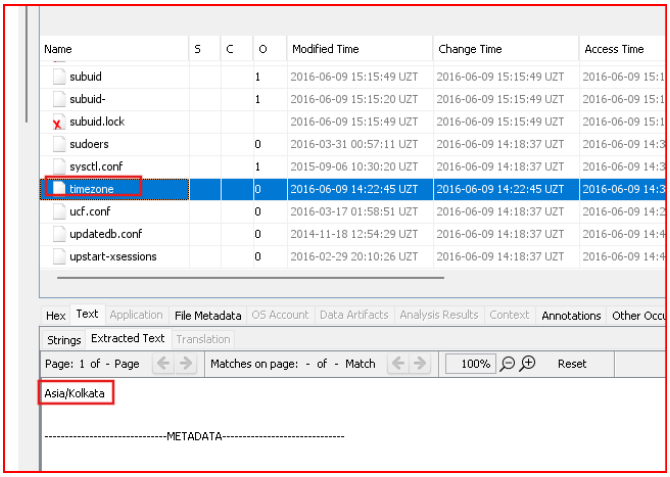
hosts content



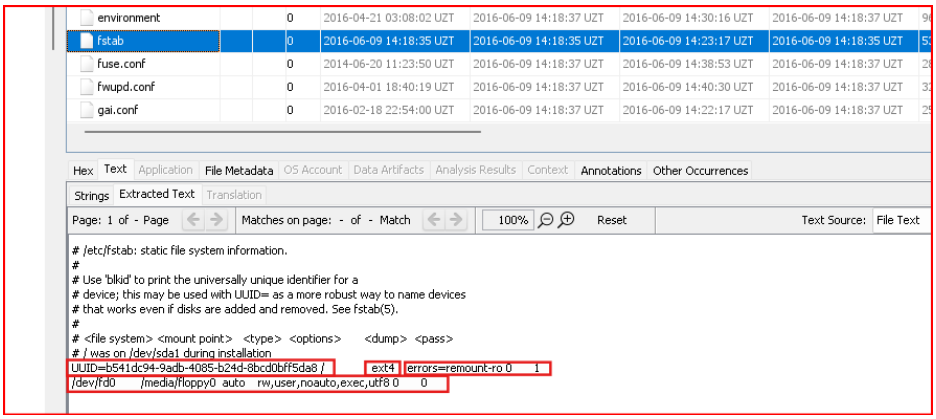
Operating System Information artifacts



timezone file



fstab content



When was the system installed?

The earliest files written by the installer give the answer: fstab at 14:18:35, hostname and hosts at 14:22:09 and timezone at 14:22:45, all on 9 June 2016 (UZT). These agree with the file system's "last checked" time of 14:18:26, which is when the file system was created. The system was therefore installed on 9 June 2016 (Moderate confidence). The April 2016 dates on lsb-release and issue are package build dates and not installation dates. The installer logs in /var/log/installer/ were not examined.

Following one file through three layers: /etc/crontab

/etc/crontab was chosen because scheduled tasks are a classic place to look for persistence or automated data movement.

It was used to demonstrate that a file is really three separate things: **a name in a directory, an inode holding metadata, and data blocks holding content.**



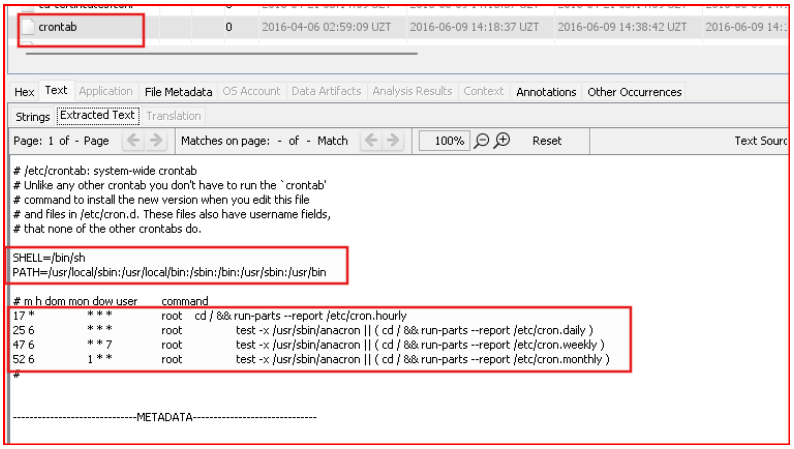
The file was first examined in Autopsy through its Text, Hex and File Metadata views. The same file was then reached with the Sleuth Kit alone. ifind converted the path into inode 127733. istat printed that inode's metadata, which matched Autopsy exactly. icat extracted the content, and certutil gave it an MD5 of 8f111d100ea459f68d333d63a8ef2205, equal to the value Autopsy had calculated. Three independent steps converged on one file.

The file is **722 bytes**, owned by root, with **one data block (589839).** Its content-modified time is 6 April 2016, while its inode was changed and created on **9 June 2016** at **14:18:37**. That gap of two months shows the content came from the software package and the inode was created at installation.

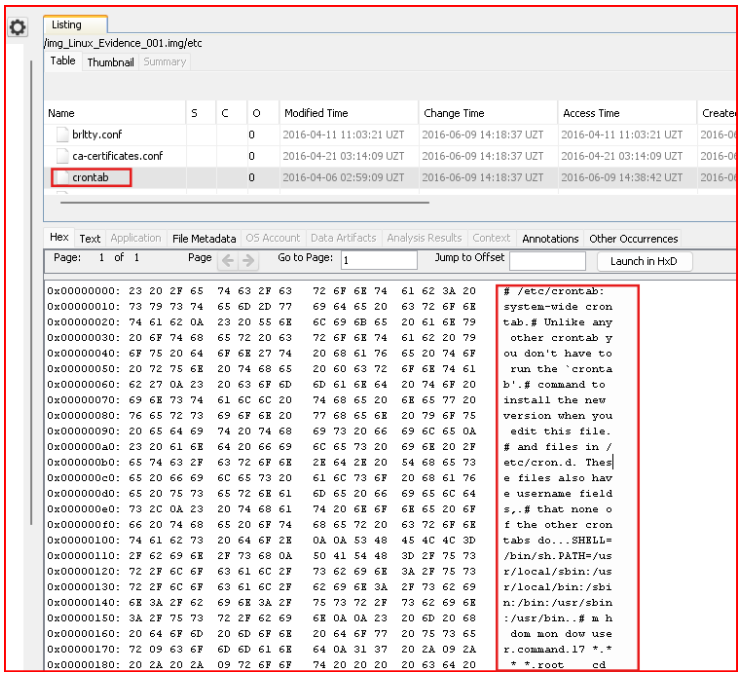
The content itself is the **standard Ubuntu configuration:** a SHELL line, a default PATH, and four root jobs that run the hourly, daily, weekly and monthly cron directories, the last three guarded by an anacron test. The file shows no sign of scheduled exfiltration or persistence. Other cron locations (/etc/cron.d, /var/spool/cron) were not examined, so this conclusion covers this file only.

Note: The inode sits in block group 16 and the data in the block range of group 18, which is consistent with how Ext4 flex block groups place metadata and data apart.

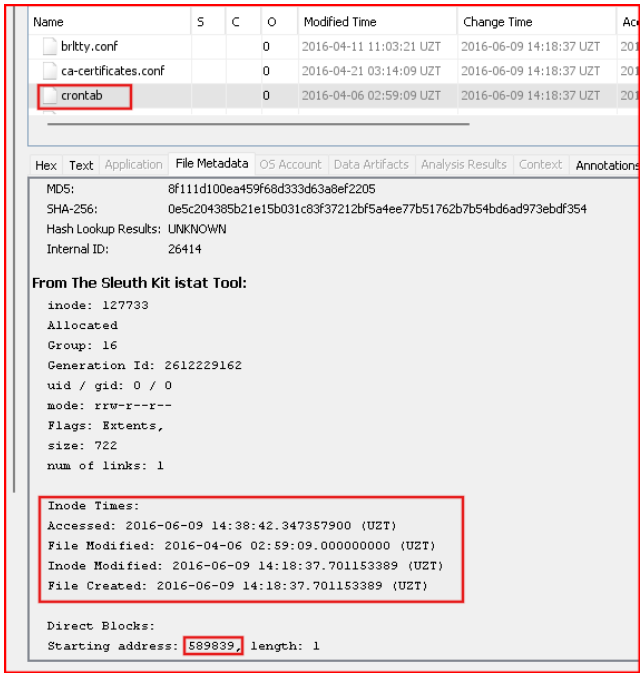
crontab, Text tab



crontab, Hex tab



crontab, File Metadata tab



ifind output

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>ifind -n etc/crontab "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img"
127733
```

istat and icat output

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img" 127733
inode: 127733
Allocated
Group: 16
Generation Id: 2612229162
uid / gid: 0 / 0
mode: rrw-r--r--
Flags: Extents,
size: 722
num of links: 1

Inode Times:
Accessed:       2016-06-09 14:38:42.347357900 (West Asia Standard Time)
File Modified:  2016-04-06 02:59:09.000000000 (West Asia Standard Time)
Inode Modified: 2016-06-09 14:18:37.701153389 (West Asia Standard Time)
File Created:   2016-06-09 14:18:37.701153389 (West Asia Standard Time)

Direct Blocks:
589839

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>icat "E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img" 127733
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab'
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin

# m h dom mon dow user  command
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily )
47 6    * * 7   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.weekly )
52 6    1 * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.monthly )
#
```

certutil MD5 of the icat output

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>certutil -hashfile crontab_icat.txt MD5
MD5 hash of crontab_icat.txt:
8f111d100ea459f68d333d63a8ef2205
CertUtil: -hashfile command completed successfully.

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>_
```

The file of interest: HideMarkedCols.xls

/home/roger/Documents/HideMarkedCols.xls is the second file the scenario names. Autopsy and istat agree that it is **inode 46066 in block group 5.** It is **49,664 bytes**, owned by **uid 1000**, and stored in **13 blocks (776861 to 776873)**. Its first bytes, **D0 CF 11 E0 A1 B1 1A E1,** are the signatures of a legacy **Microsoft Office container**. Its MD5 is **51ae5c0f8442964147f11c04676a411e**. No hash set that was loaded recognized it.

The timestamps tell the key story. Its content-modified time is **28 September 2011**, nearly five years before the system existed. Its creation time on this volume is **9 June 2016 at 15:06:45,** and **its inode was last changed at 15:42:05**.

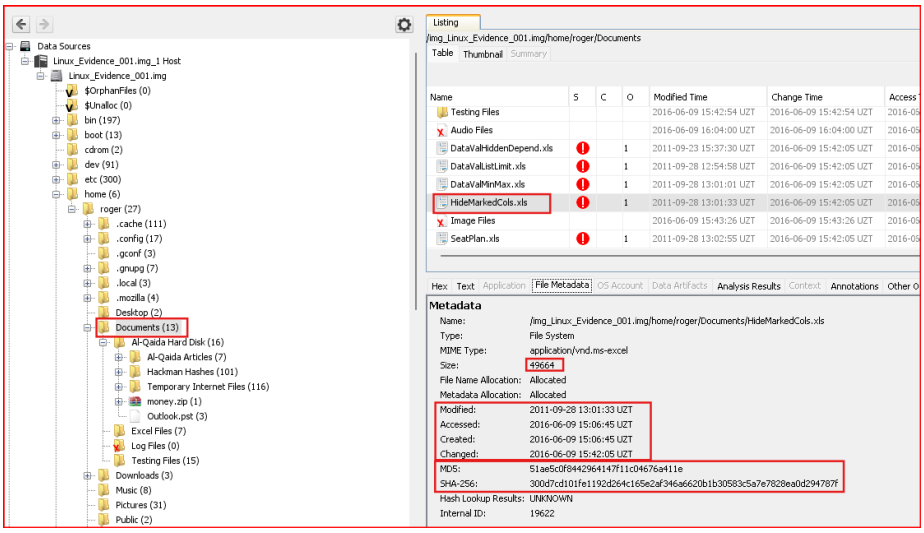
A file cannot have been last edited five years before the volume it sits on was created, unless it arrived from somewhere else with its original modified date preserved, as happens when files are copied or extracted from an archive in a way that keeps timestamps. So, the file was introduced onto this system and not created on it (High confidence). Its sibling workbooks in the same folder share the **2011 date pattern** and the same **15:42:05** change time, which suggests they were copied or handled together (Moderate).

Note: The filename is recorded as an observation only; no inference about intent or content is drawn from it.

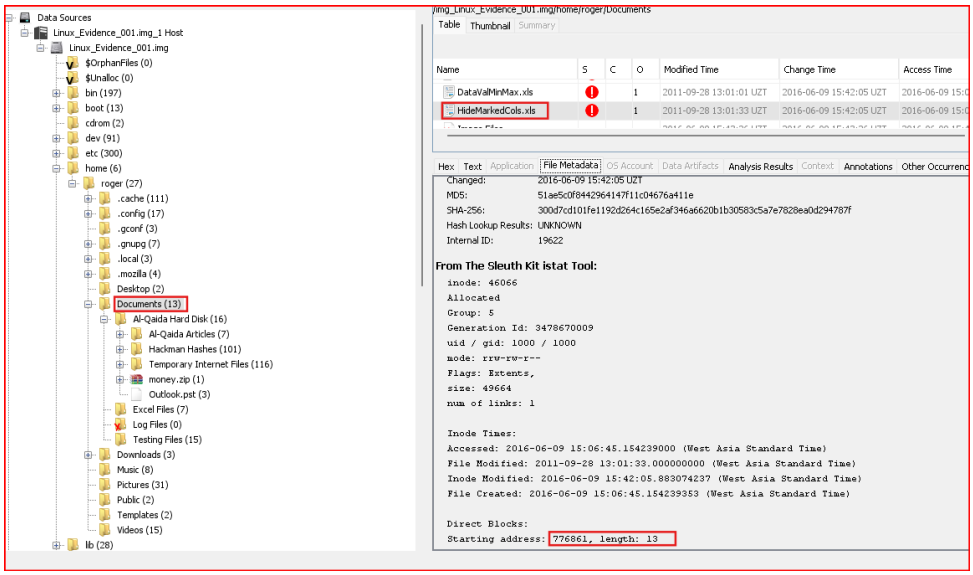
To verify the hash, the file was exported from Autopsy, and a second copy was produced with icat. All three MD5 values were identical. The exports were only hashed, never opened, to avoid triggering any macro or altering them.

The arithmetic of its storage became important: 49,664 bytes in 13 blocks of 4,096 (53,248 bytes) leaves 3,584 bytes of unused space at the end of the last block, known as file slack.

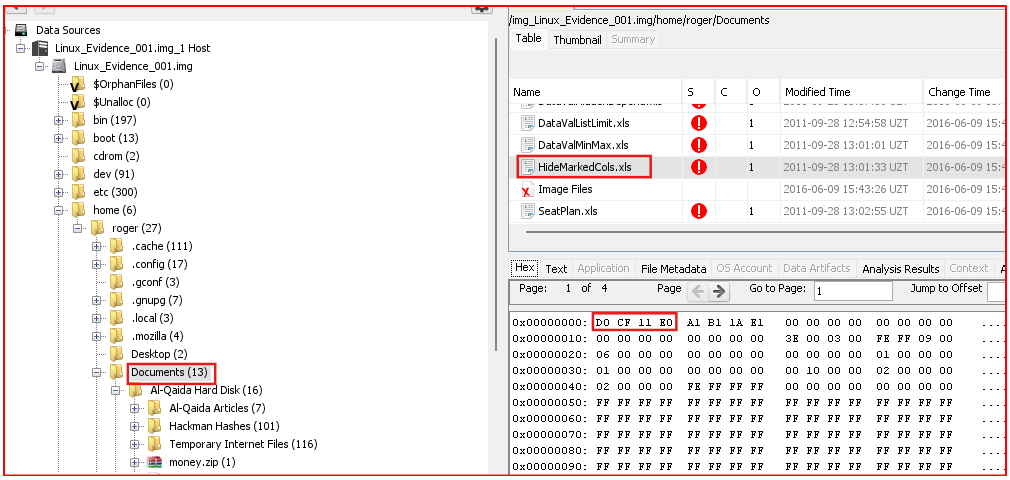
HideMarkedCols.xls, File Metadata (Autopsy)



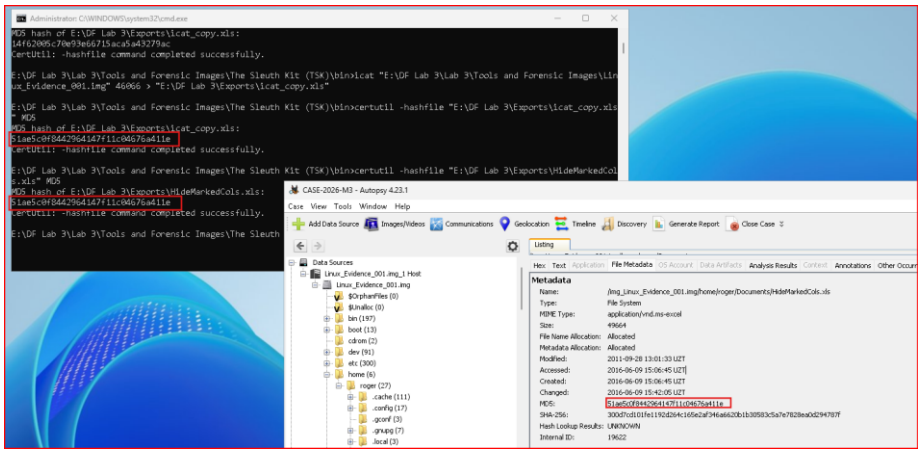
HideMarkedCols.xls, File Metadata with the istat section



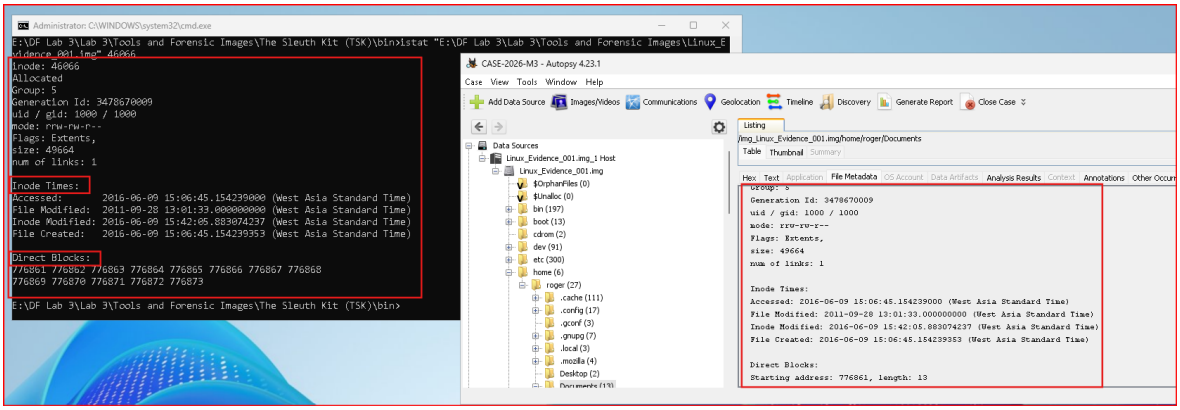
HideMarkedCols.xls, Hex tab showing D0 CF 11 E0



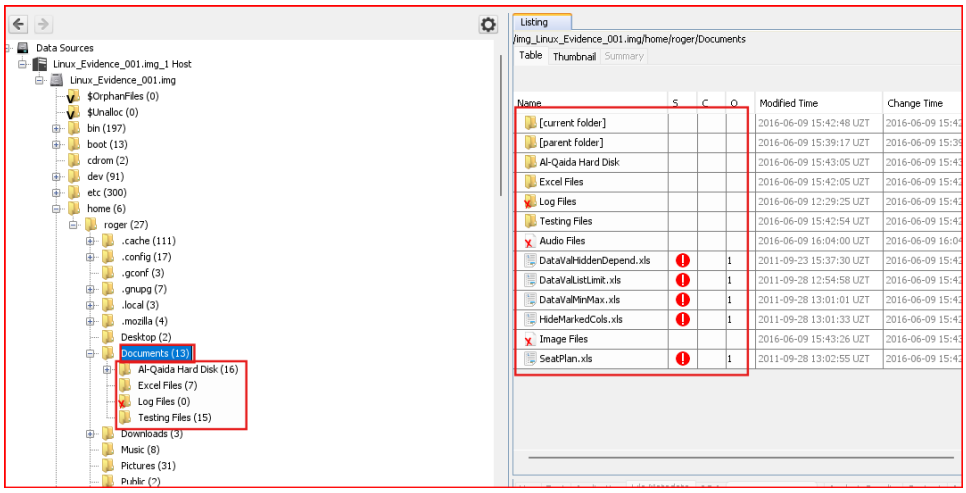
Extract File(s) to Exports, and certutil MD5 of both copies beside the Autopsy MD5



istat output beside the Autopsy metadata



Documents folder listing showing deleted markers and flagged workbooks



A system that did not stop cleanly

fsstat records the last **mount at 15:36:52,** yet the file system contains events after that moment. HideMarkedCols.xls's inode changed at 15:42:05, and directory access times run as late as 15:45:57. A mount session that began at 15:36:52 therefore carried on for at least nine more minutes without the superblock being rewritten. This fits the "Needs Recovery" flag and the orphan-inode list in the superblock, and it contradicts fsstat's statement that the file system was unmounted properly.

Note: The most reasonable reading is that the system was stopped abruptly (Moderate). The exact stop time cannot be fixed from this evidence. Later extended this observation (Section 10).

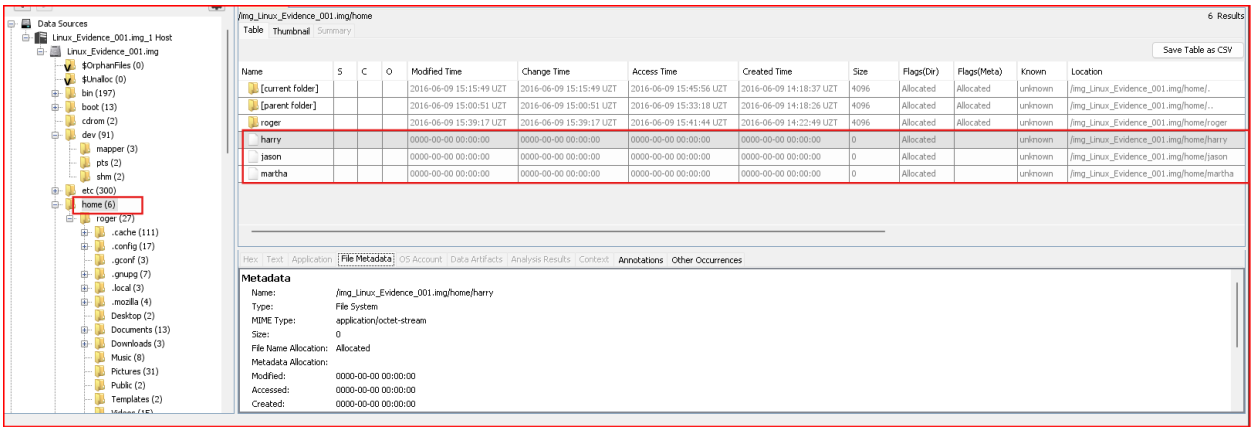
The image is incomplete

The most important limitation of EV-001 emerged in Autopsy's /home listing. The image holds blocks 0 to 1,012,044, while the file system declares blocks 0 to 1,291,263. Inode tables for block groups 32 to 39 begin at block 1,048,582 or higher**, which is outside the image**.

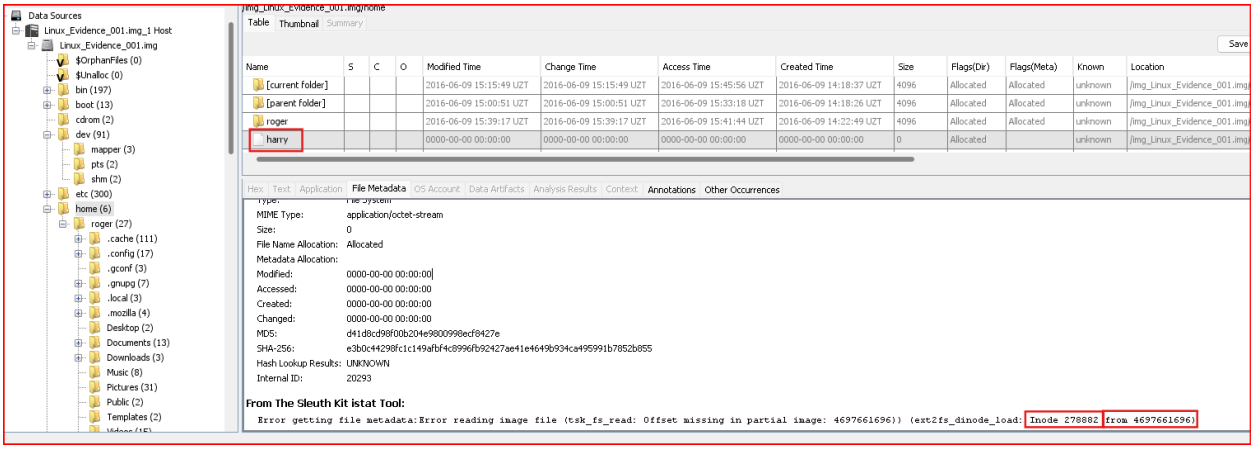
Several home directories, including harry, jason and martha, appear in the directory with timestamps of 0000-00-00. Their names survive in the directory entries, but their inode records are physically absent. An attempt to read harry's inode produced **"Offset missing in partial image".**

This is a limitation of the evidence, not a sign of deletion. Examining those users requires a complete image of the source disk. This limit also explains the failures of some inode-level commands later.

/home listing showing zero timestamps



File Metadata for harry (inode 278882)



Understanding the Windows Images (EV-002 & EV-003)

Finding the layout

The same first question was asked of the Windows images, and it produced two different answers. mmls on EV-002 returned "**Cannot determine partition type**", the same message as for EV-001. fsstat then ran successfully with no offset and **identified NTFS.**

WinHex confirmed it from raw bytes: at **offset 0 it shows EB 52 90** followed by the characters NTFS, the boot sector of an NTFS volume at the very first sector.

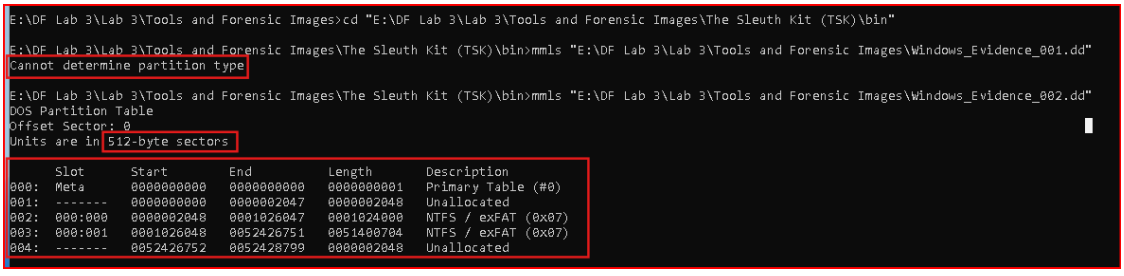
Note: EV-002 is therefore bare volume and needs no offset.

EV-003 behaved differently. mmls found a DOS partition table, and WinHex showed the boot signature **55 AA at offsets 510** and **511** together with the standard boot-code messages. The table is as follows.

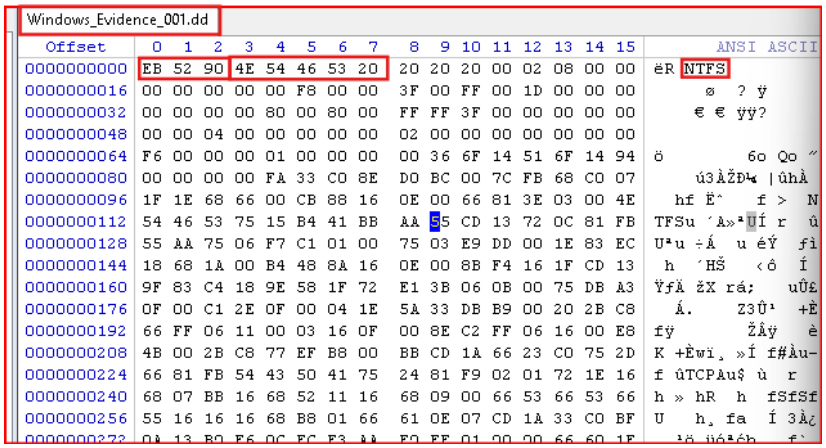
Slot	Start sector	End sector	Length (sectors)	Type
Unallocated	0	2,047	2,048	n/a
Partition 1	2,048	1,026,047	1,024,000	NTFS (0x07)
Partition 2	1,026,048	52,426,751	51,400,704	NTFS (0x07)
Unallocated	52,426,752	52,428,799	2,048	n/a

Partition 1 is 500 MiB and begins on a 1 MiB boundary. Partition 2 is about 24.5 GiB. A further 1 MiB is left unallocated at the end. Together the partitions and the unallocated areas account for all **52,428,800 sectors**. Any command aimed at EV-003 needs -o 2048 or -o 1026048 to reach a file system.

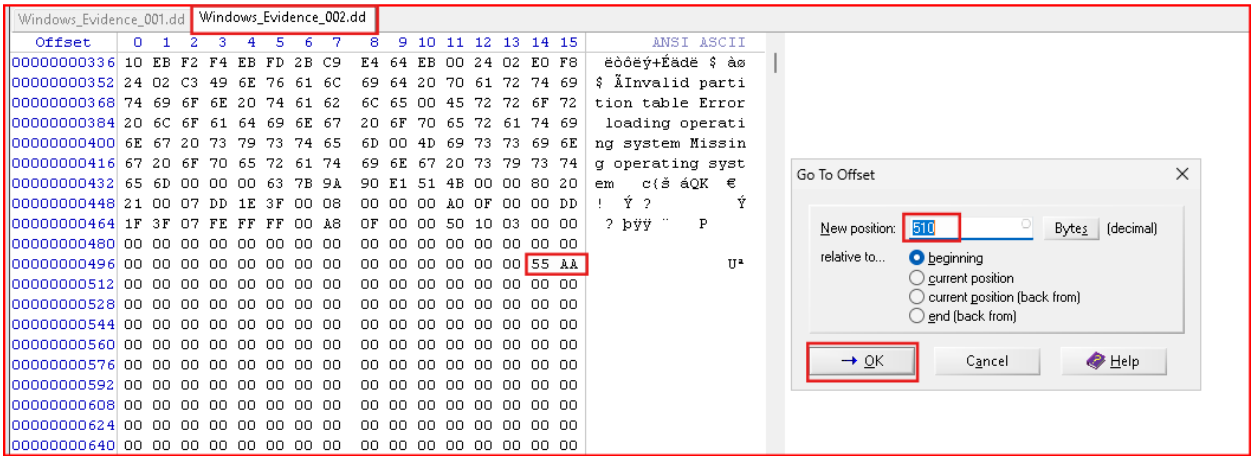
mmls output for both images



WinHex, Windows_Evidence_001.dd, offset 0, NTFS boot sector



WinHex, Windows_Evidence_002.dd, Go to Offset 510, bytes 55 AA



Describing the three NTFS volumes

fsstat was then run on each of the three volumes. All three use **512-byte sectors, 4,096-byte clusters, 1,024-byte MFT** records and the root directory at MFT entry 5. They differ in size and serial number:

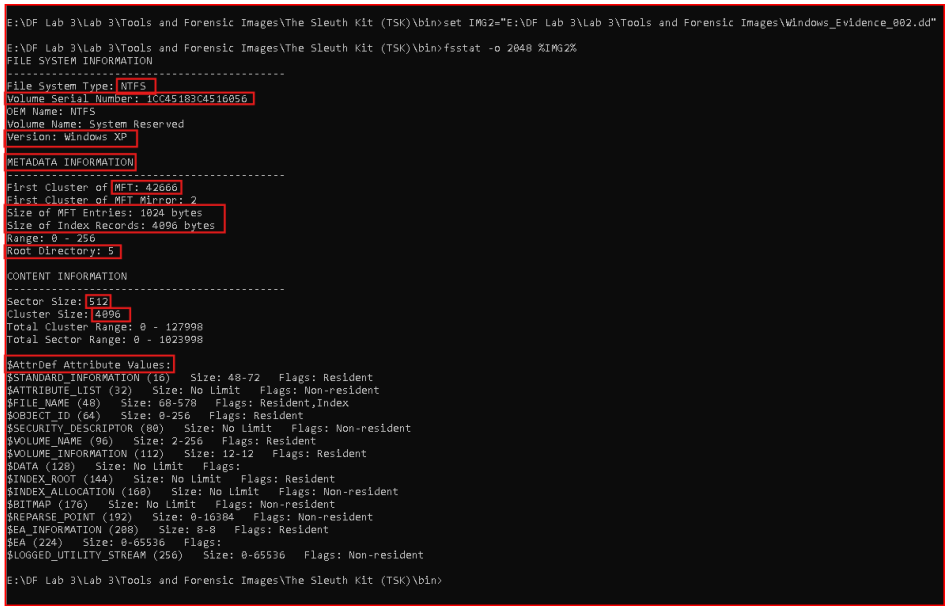
EV-002's volume is named "Evidence" (serial 94146F51146F3600) and spans 524,287 clusters.

EV-003's first partition is named "System Reserved" (serial 1CC45183C4516056) and its second (serial 163E52F43E52CBFD) has no name. The label "Windows XP" that fsstat print is the NTFS version (3.1) and says nothing about the installed operating system. The attribute-definition table is identical on all three volumes, as expected.

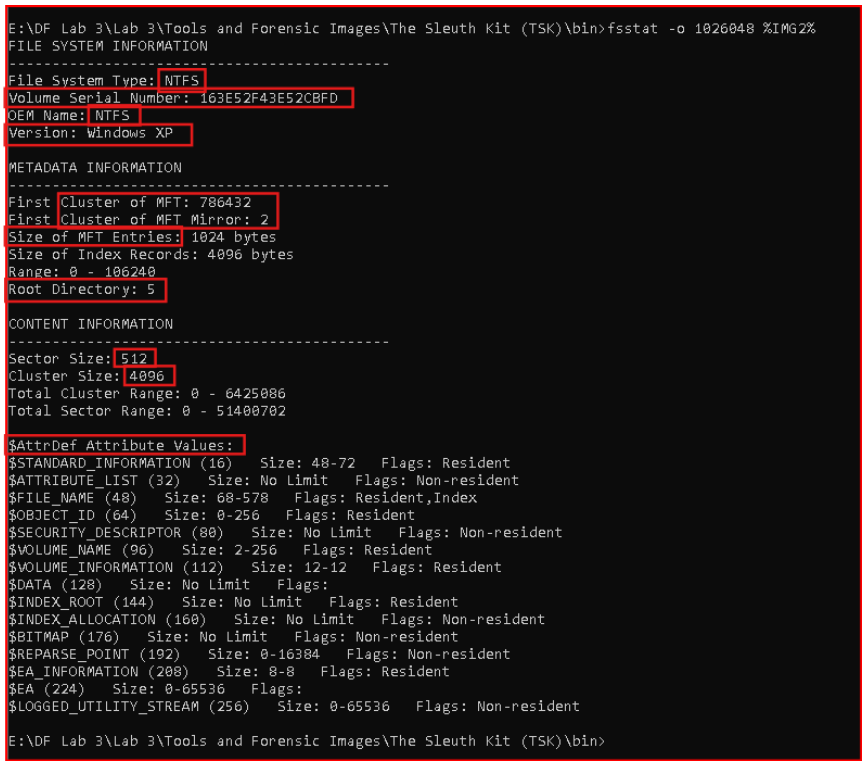
Each file system's sector range is one sector shorter than its container, which is consistent with NTFS keeping a backup boot sector in the final sector (Moderate).

For EV-002 the size arithmetic was checked: **2,147,483,648 bytes is 4,194,304 sectors**, and the file system's range ends at **sector 4,194,302. Its 524,287** clusters of **4,096 bytes** total 2,147,479,552 bytes. Nothing is missing. EV-002 holds the whole volume, unlike EV-001.

fsstat -o 2048, EV-003 partition 1



fsstat -o 1026048, EV-003 partition 2



fsstat -f ntfs, EV-002

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>fsstat -f ntfs %IMG%
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: 94146F51146F3600
OEM Name: NTFS
Volume Name: Evidence
Version: Windows XP

METADATA INFORMATION
--------------------------------------------
First Cluster of MFT: 262144
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 1280
Root Directory: 5

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 524286
Total Sector Range: 0 - 4194302

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)   Size: 48-72   Flags: Resident
$ATTRIBUTE_LIST (32)   Size: No Limit   Flags: Non-resident
$FILE_NAME (48)   Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)   Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit   Flags: Non-resident
$VOLUME_NAME (96)   Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)   Size: No Limit   Flags:
$INDEX_ROOT (144)   Size: No Limit   Flags: Resident
$INDEX_ALLOCATION (160)   Size: No Limit   Flags: Non-resident
$BITMAP (176)   Size: No Limit   Flags: Non-resident
$REPARSE_POINT (192)   Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)   Size: 8-8   Flags: Resident
$EA (224)   Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256)   Size: 0-65536   Flags: Non-resident
```

img_stat, EV-002

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>img_stat %IMG%
IMAGE FILE INFORMATION
--------------------------------------------
Image Type: raw

Size in bytes: 2147483648
Sector size:    512

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>
```

Reading the NTFS system files

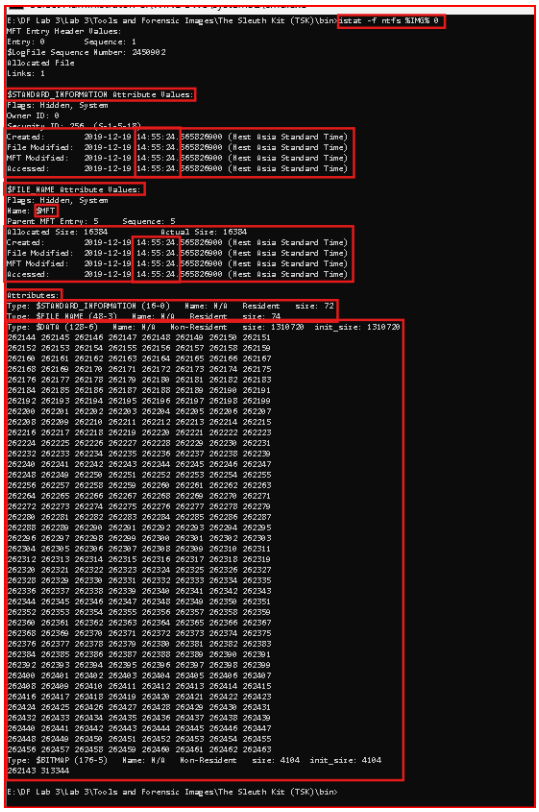
NTFS keeps its own bookkeeping in special files at the start of the Master File Table (MFT). Entries 0 to 9 of EV-002 were each examined with istat. All ten are allocated. Entries 0–4 and 6–9 carry the Hidden and System flags, and entry 5 is the root directory. **The key structures were:**

- $MFT (entry 0): 1,310,720 bytes in 320 clusters starting at cluster 262,144.
- $MFTMirr (entry 1): a 4,096-byte backup at cluster 2.
- $LogFile (entry 2): a 6,356,992-byte transaction log.
- $Volume (entry 3): holds the 16-byte volume name, which is the eight UTF-16 characters of "Evidence".
- $AttrDef (entry 4): 2,560 bytes at cluster 35.
- Root directory (entry 5): a directory index at cluster 36.
- $Bitmap (entry 6): 65,536 bytes, which is 524,288 bits and covers all 524,287 clusters.
- $Boot (entry 7): clusters 0 and 1, matching the boot sector seen in WinHex.
- $BadClus (entry 8): a volume-sized sparse stream with an initialised size of zero, meaning no bad clusters are recorded.
- $Secure (entry 9): the security-descriptor store.

The values were compared against fsstat and WinHex. The $MFT start cluster agrees with fsstat, the $Boot location agrees with WinHex, and the $BadClus size equals exactly 524,287 clusters of 4,096 bytes. The system files hold standard NTFS structures (High confidence).

Their timestamps are revealing. Entries 0–4 and 6–9 all carry one identical value to the nanosecond, 2019-12-19 09:55:24.566 UTC, in both of their timestamp attribute sets ($STANDARD_INFORMATION and $FILE_NAME). This indicates creation in a single operation, most probably formatting, which would make 09:55:24 UTC the creation instant of the volume (Moderate). The root directory was modified about 33 seconds later, at 09:55:57 UTC.

istat entry 0 ($MFT)



istat entry 1 ($MFTMirr)

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat -f ntfs %IMG% 1
MFT Entry Header Values:
Entry: 1        Sequence: 1
$LogFile Sequence Number: 2118363
Allocated File
Links: 1

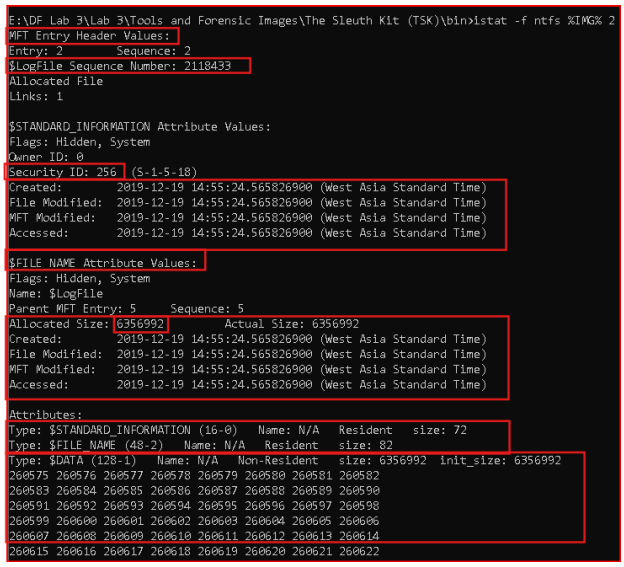
$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-18)
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $MFTMirr
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 4096    Actual Size: 4096
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

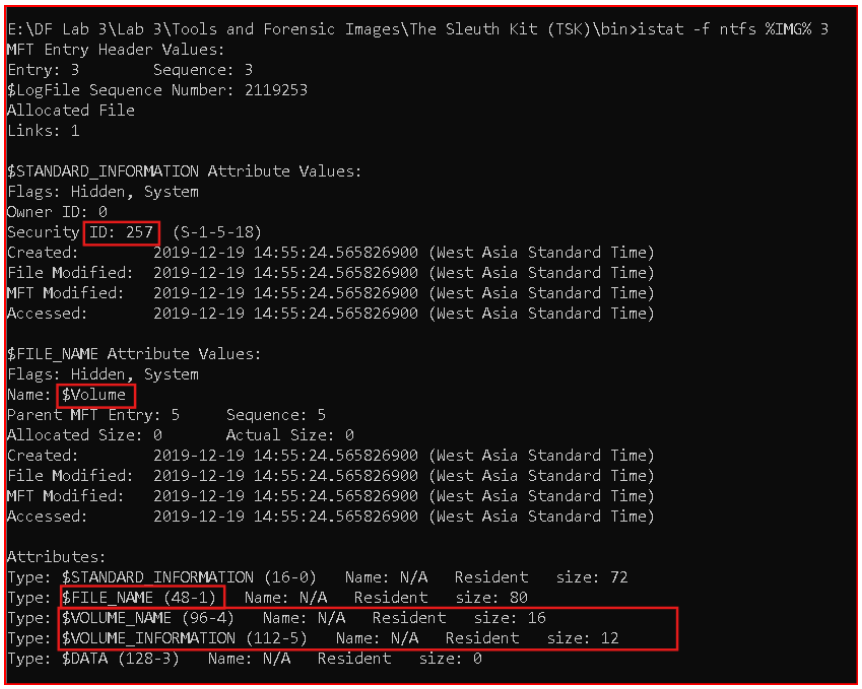
Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 82
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 4096  init_size: 4096
2

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>
```

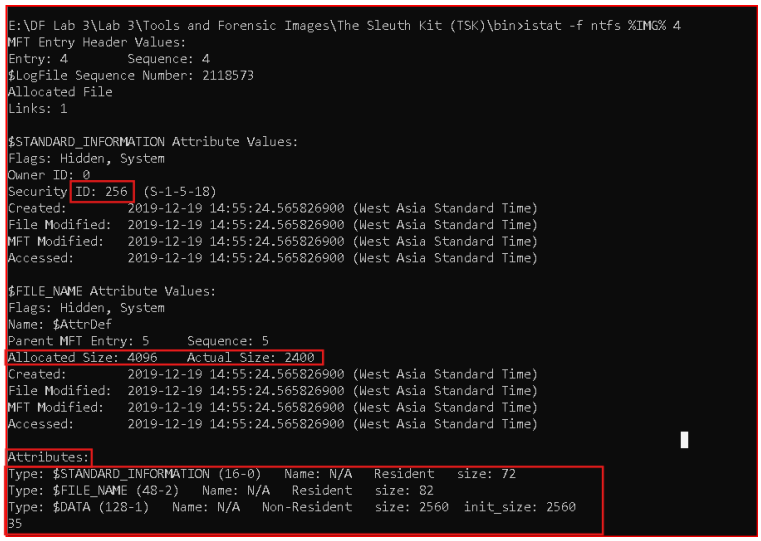
istat entry 2 ($LogFile)



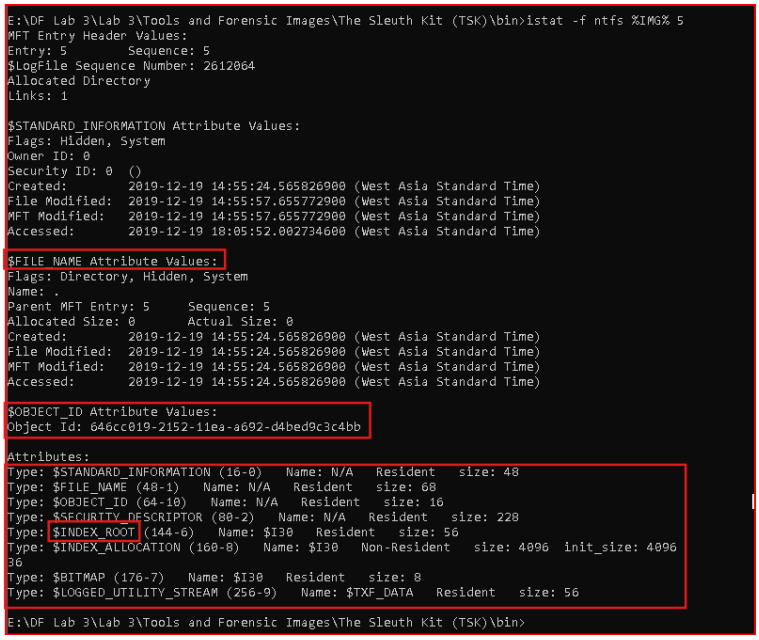
istat entry 3 ($Volume)



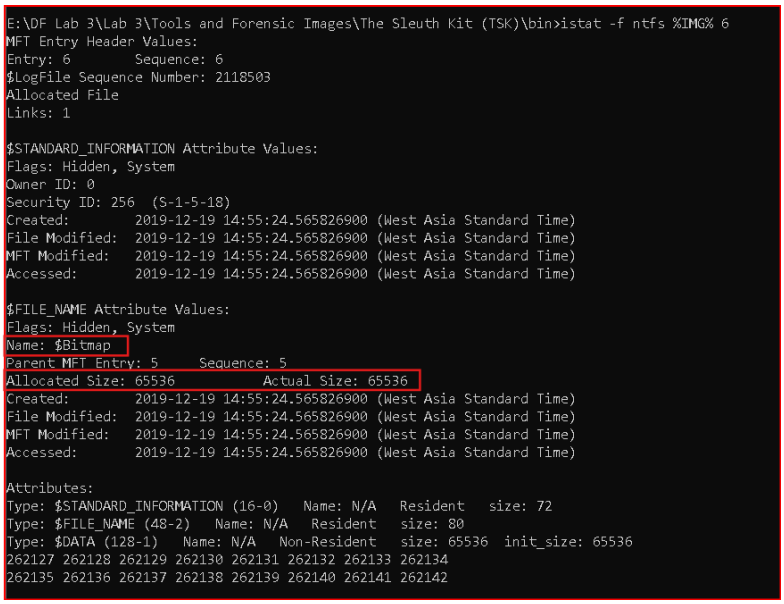
istat entry 4 ($AttrDef)



istat entry 5 (root directory)



istat entry 6 ($Bitmap)



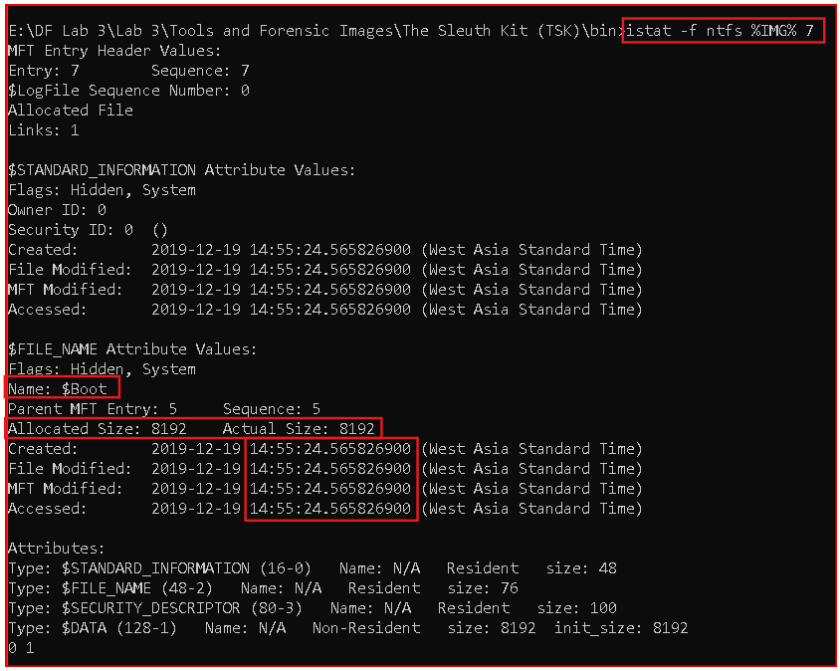
```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat -f ntfs %IMG% 6
MFT Entry Header Values:
Entry: 6        Sequence: 6
$LogFile Sequence Number: 2118503
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-18)
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Bitmap
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 65536           Actual Size: 65536
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 80
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 65536  init_size: 65536
262127 262128 262129 262130 262131 262132 262133 262134
262135 262136 262137 262138 262139 262140 262141 262142
```

istat entry 7 ($Boot)



```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat -f ntfs %IMG% 7
MFT Entry Header Values:
Entry: 7        Sequence: 7
$LogFile Sequence Number: 0
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0  ()
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Boot
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 8192    Actual Size: 8192
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 48
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 76
Type: $SECURITY_DESCRIPTOR (80-3)   Name: N/A   Resident   size: 100
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 8192  init_size: 8192
0 1
```

istat entry 8 ($BadClus)

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat -f ntfs %IMG% 8
MFT Entry Header Values:
Entry: 8        Sequence: 8
$LogFile Sequence Number: 2118643
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-18)
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $BadClus
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 0       Actual Size: 0
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-3)   Name: N/A   Resident   size: 82
Type: $DATA (128-2)   Name: N/A   Resident   size: 0
Type: $DATA (128-1)   Name: $Bad   Non-Resident   size: 2147479552  init_size: 0

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>
```

istat entry 9 ($Secure)

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat -f ntfs %IMG% 9
MFT Entry Header Values:
Entry: 9        Sequence: 9
$LogFile Sequence Number: 2528320
Allocated File
Links: 1

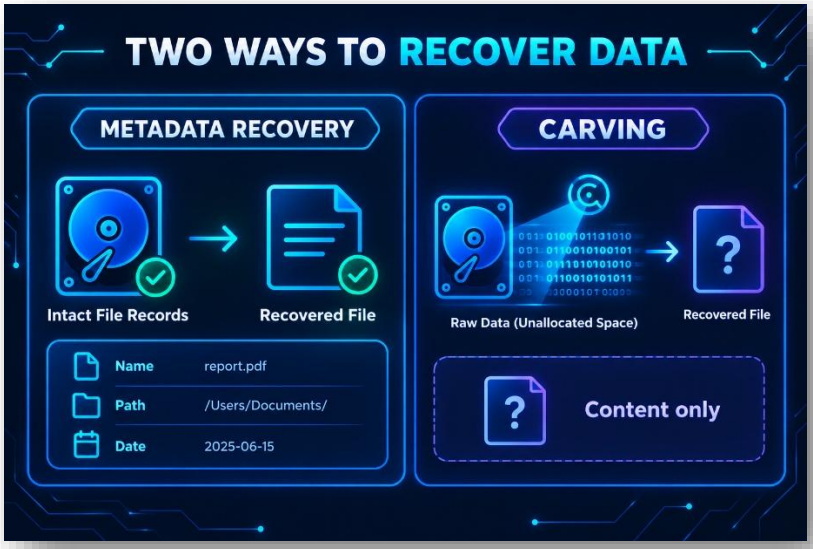
$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 257  (S-1-5-18)
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System, Index View
Name: $Secure
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 0       Actual Size: 0
Created:        2019-12-19 14:55:24.565826900 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:24.565826900 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:24.565826900 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:24.565826900 (West Asia Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-7)   Name: N/A   Resident   size: 80
Type: $DATA (128-8)   Name: $SDS   Non-Resident   size: 264468  init_size: 264468
314062 314063 314064 314065 314066 314067 314068 314069
314070 314071 314072 314073 314074 314075 314076 314077
314078 314079 314080 314081 314082 314083 314084 314085
314086 314087 314088 314089 314090 314091 314092 314093
314094 314095 314096 314097 314098 314099 314100 314101
314102 314103 314104 314105 314106 314107 314108 314109
314110 314111 314112 314113 314114 314115 314116 314117
314118 314119 314120 314121 314122 314123 314124 314125
314126
Type: $INDEX_ROOT (144-11)   Name: $SDH   Resident   size: 56
Type: $INDEX_ROOT (144-14)   Name: $SII   Resident   size: 56
Type: $INDEX_ALLOCATION (160-9)   Name: $SDH   Non-Resident   size: 4096  init_size: 4096
315423
Type: $INDEX_ALLOCATION (160-12)   Name: $SII   Non-Resident   size: 4096  init_size: 4096
315424
Type: $BITMAP (176-10)   Name: $SDH   Resident   size: 8
Type: $BITMAP (176-13)   Name: $SII   Resident   size: 8
```

Recovering Deleted and Hidden Data

Recovery was approached in two fundamentally different ways, and the contrast between them is one of the main lessons of the module. **Metadata-based recovery** uses the file system's own leftover records and returns names, paths and timestamps along with the content. **Carving** ignores the file system entirely and searches the raw bytes for known file headers, which recover content only, with no names or dates.



Cataloguing the deleted entries on EV-002

fls -r -p listed every entry on EV-002 (1,327 lines), and the same command with -d listed only deleted entries (79 lines). Counting the markers in those 79 lines gave this breakdown.

Category	Count
Deleted files with an MFT entry and a data stream	22
Zone.Identifier alternate data streams of deleted files	6
Directory-entry lines with MFT reference 0 (no linked metadata)	51
Total	79

The 22 files are **inodes 81–83** (three image files), **93** (a text file), **94** (a presentation), **95** and **96** (two Word documents), **1189–1199** (eleven audio files) and **1213, 1214, 1225** and

1226 (four text files). ils -r independently listed the same **22 inodes**. The **51** reference-0 lines are directory entries whose metadata has been overwritten or reused, so they cannot be recovered this way. They are not 51 distinct files: three carry the same names as recoverable files (Word_Doc1.docx and two audio tracks). No name-by-name reconciliation was carried out.

The Zone. Identifier streams are small named streams that Windows attaches to files downloaded from elsewhere. Six of the deleted files carry one (**inodes 81, 82, 83, 94, 95 and 96).** Only the stream of **inode 96** was examined at metadata level, and no stream content was read.

fls commands writing fls_all.txt and fls_deleted.txt

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>fls -f ntfs -r -p %IMG% > "E:\DF Lab 3\Exports\fls_all.txt"

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>fls -f ntfs -r -p -d  %IMG% > "E:\DF Lab 3\Exports\fls_deleted.txt"
```

fls_deleted.txt open in Notepad



line counts of both listings

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>find /c /v "" "E:\DF Lab 3\Exports\fls_all.txt"

---------- E:\DF LAB 3\EXPORTS\FLS_ALL.TXT: 1327

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>find /c /v "" "E:\DF Lab 3\Exports\fls_deleted.txt"

---------- E:\DF LAB 3\EXPORTS\FLS_DELETED.TXT: 79

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>_
```

```
 Directory of E:\DF Lab 3\Exports

10/03/2026  03:46 PM            28,160 inode96_icat.docx
               1 File(s)         28,160 bytes
               0 Dir(s)  28,256,108,544 bytes free

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>find /c "-128-1:" "E:\DF Lab 3\Exports\fls_deleted.txt"

---------- E:\DF LAB 3\EXPORTS\FLS_DELETED.TXT: 22

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>find /c "-128-3:" "E:\DF Lab 3\Exports\fls_deleted.txt"

---------- E:\DF LAB 3\EXPORTS\FLS_DELETED.TXT: 6

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>find /c "* 0:" "E:\DF Lab 3\Exports\fls_deleted.txt"

---------- E:\DF LAB 3\EXPORTS\FLS_DELETED.TXT: 51

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>
```

ils -r output

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>ils -r %IMG%
class|host|device|start_time
ils|unknown||1791023940
st_ino|st_alloc|st_uid|st_gid|st_mtime|st_atime|st_ctime|st_crtime|st_mode|st_nlink|st_size
81|f|0|0|1556712888|1576760743|1556780341|1576749349|777|1|19575
82|f|0|0|1556713107|1576760743|1556780341|1576749349|777|1|31740
83|f|0|0|1556712909|1576760743|1556780341|1576749349|777|1|15798
93|f|0|0|1479206858|1576749349|1576746651|1576749349|777|1|446
94|f|0|0|1479126739|1576749349|1576746651|1576749349|777|1|81920
95|f|0|0|1466138830|1576749349|1576746651|1576749349|777|1|28160
96|f|0|0|1466058746|1576749349|1576746651|1576749349|777|1|28160
1189|f|0|0|1515828310|1576760727|1576746861|1576749354|777|1|6852177
1190|f|0|0|1515779012|1576760727|1576746861|1576749355|777|1|8366232
1191|f|0|0|1515779024|1576760727|1576746861|1576749355|777|1|7024583
1192|f|0|0|1515779032|1576760727|1576746861|1576749355|777|1|8883453
1193|f|0|0|1515779046|1576760727|1576746861|1576749355|777|1|7934690
1194|f|0|0|1515779062|1576760728|1576746861|1576749355|777|1|8691195
1195|f|0|0|1515779076|1576760728|1576746862|1576749355|777|1|11987383
1196|f|0|0|1515779086|1576760728|1576746862|1576749355|777|1|8276372
1197|f|0|0|1515779096|1576760728|1576746862|1576749355|777|1|9001197
1198|f|0|0|1515779108|1576760728|1576746862|1576749356|777|1|10768453
1199|f|0|0|1515779120|1576760728|1576746862|1576749356|777|1|5942069
1213|f|0|0|1556778286|1576749357|1556780336|1576749357|777|1|7943
1214|f|0|0|1556778302|1576749357|1556780336|1576749357|777|1|6551
1225|f|0|0|1556778255|1576749357|1556780336|1576749357|777|1|5756
1226|f|0|0|1556778272|1576749357|1556780336|1576749357|777|1|4635

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>_
```

Following one deleted file end to end: inode 96

Inode 96 was followed through four Sleuth Kit tools. ffind resolved the inode to the path /Other Files/Word_Doc1.docx, with an asterisk marking it as deleted.

istat showed the metadata: the entry is not allocated, it has one link, and its parent is entry **85**. The data stream is non-resident, **28,160 bytes** in **seven clusters (346,899 to 346,905),** which gives **28,672 bytes allocated** and **512 bytes of slack**. The security identifier ends in the relative identifier **1002**, which denotes an account created on the source system. The account name was not resolved.

icat extracted the content to inode96_icat.docx: 28,160 bytes, the same size as the recorded data stream, with an MD5 of eabeb38580705f67e46da01b8d311897. blkstat confirmed that cluster 346,899 is not allocated, so the file's first cluster has not been given to another file. Together these show that the deleted entry's content is likely intact (Moderate).

The timestamps of inode 96 are unusual and recur throughout the evidence.

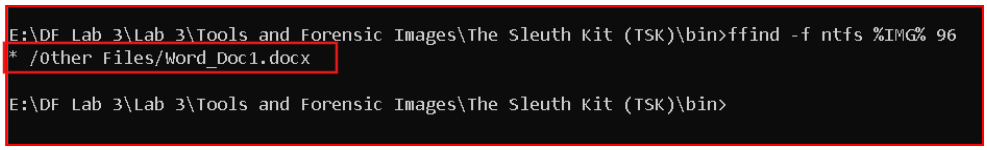
Its content was last modified on 2016-06-16 06:32:26 UTC.

Its MFT record was modified on 2019-12-19 at 09:10:51 UTC.

It was created on 2019-12-19 at 09:55:49 UTC, 25 seconds after the volume's system files were written.

Note: The MFT-modified time is therefore earlier than the creation time, and earlier even than the volume's own creation.

ffind output for inode 96



istat output for inode 96

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>istat -f ntfs %IMG% 96
MFT Entry Header Values:
Entry: 96        Sequence: 2
$LogFile Sequence Number: 2609486
Not Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 265  (S-1-5-21-3041995720-2451581635-492656684-1002)
Created:        2019-12-19 14:55:49.170990000 (West Asia Standard Time)
File Modified:  2016-06-16 11:32:26.000000000 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:10:51.774038100 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:49.181120800 (West Asia Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: Word_Doc1.docx
Parent MFT Entry: 85   Sequence: 1
Allocated Size: 28672          Actual Size: 0
Created:        2019-12-19 14:55:49.170990000 (West Asia Standard Time)
File Modified:  2019-12-19 14:55:49.170990000 (West Asia Standard Time)
MFT Modified:   2019-12-19 14:55:49.170990000 (West Asia Standard Time)
Accessed:       2019-12-19 14:55:49.170990000 (West Asia Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 94
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 28160  init_size: 28160
346899 346900 346901 346902 346903 346904 346905
Type: $DATA (128-3)   Name: Zone.Identifier   Resident   size: 534
```

icat to inode96_icat.docx and certutil MD5

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>icat -f ntfs %IMG% 96 > "E:\DF Lab 3\Exports\inode96_icat.docx"

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>certutil -hashfile "E:\DF Lab 3\Exports\inode96_icat.docx" MD5
MD5 hash of E:\DF Lab 3\Exports\inode96_icat.docx:
eabeb38580705f67e46da01b8d311897
CertUtil: -hashfile command completed successfully.
```

blkstat for cluster 346,899

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>blkstat -f ntfs %IMG% 346899
Cluster: 346899
Not Allocated

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>
```

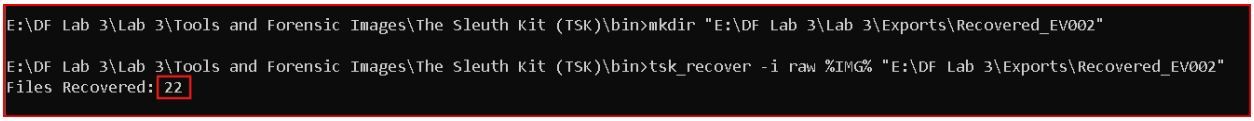
Recovering all deleted files at once

tsk_recover was run on EV-002 to recover every deleted file in one operation. The tool reported **22 files recovered**, and a recursive PowerShell count of the output folder returned 22.

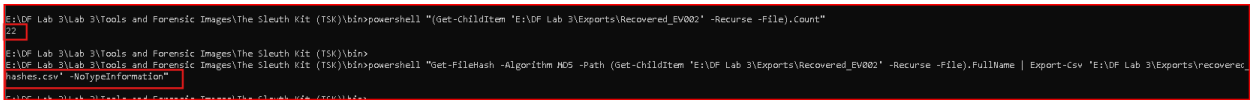
These two numbers match the **22 deleted files** with metadata that fls and ils reported, so four methods agree on the count. MD5 values of the recovered files were exported to a CSV. The six alternate streams were not counted as files. The output of tsk_recover -h was not captured, so the default scope (unallocated files) rests on documented behaviour and on the matching count.

The timestamp anomaly seen in **inode 96** was then tested across all recovered entries using the epoch values from ils -r. In **every one of the 22 entries** the MFT-modified time precedes the creation time, and creation times fall **25 to 33 seconds** after the volume was formatted. The pattern is systematic and not a quirk of one file (Moderate). Its cause was not determined (Low), and timestamp-consistency analysis is carried forward to a later module.

mkdir and tsk_recover output (Files Recovered: 22)



PowerShell recovered-file count (22) and MD5 export

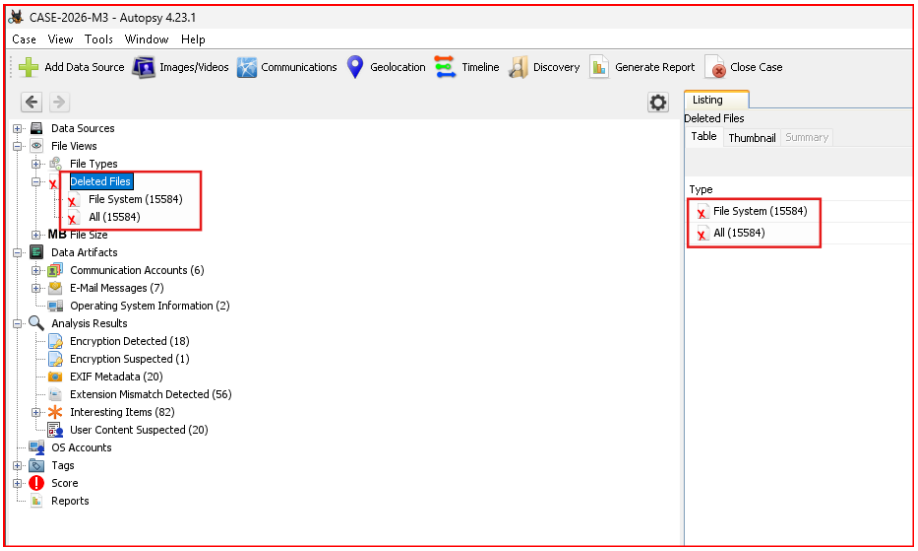


Deleted entries on the Linux image

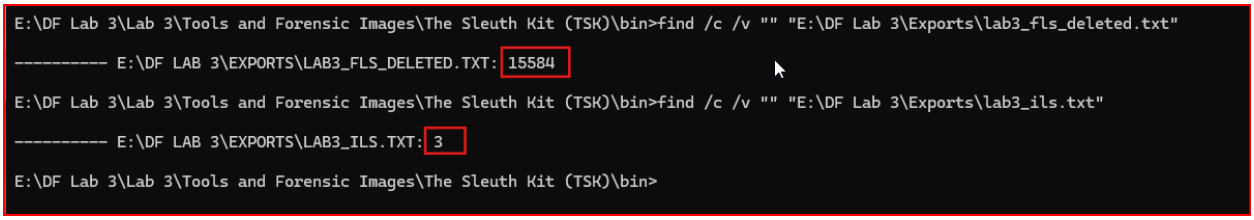
The same questions were put to EV-001. fls produced a list of **15,584** deleted entries, and Autopsy's Deleted Files view, checked under both "File System" and "All", also showed 15,584. Two tools agreed. The inode listing was a different story: ils produced only three lines. The examiner's working explanation is that the inode records lie outside the truncated image, but that remains an inference (Low).

Note: It is important to remember that a name marked as deleted is only a directory entry and does not prove that any content can be recovered.

Autopsy case window, Deleted Files showing 15584 (File System and All)]



find /c /v output



Carving the Linux image with WinHex

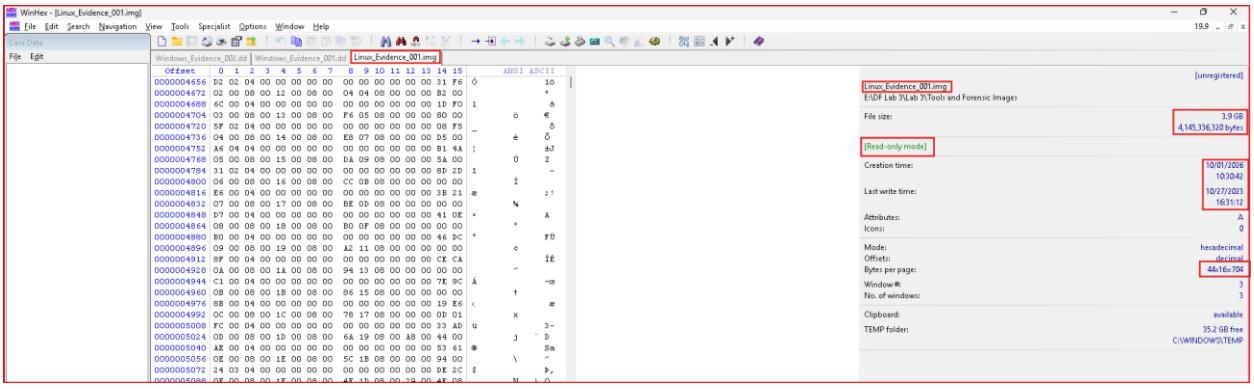
With the file system only partly usable, carving was the natural next step. First, read-only handling was re-confirmed in WinHex, whose status panel showed the read-only indicator and the exact file size of **4,145,336,320 bytes**, and the SHA-256 was re-hashed and matched.

WinHex's ***File Recovery by Type*** function was then run over the whole image (offset 0 to 4,145,336,319). The types ticked were **OLE2/MS Office, Office 2007+, Rich Text Format and PDF (Base64),** with a subfolder per file type and a new subfolder after 2,500 files. The run took about **12 minutes (02:01:33 to 02:13:21 on 4 October 2026).** The tool's own log notes that the evaluation version cannot save files larger than **200 KB**.

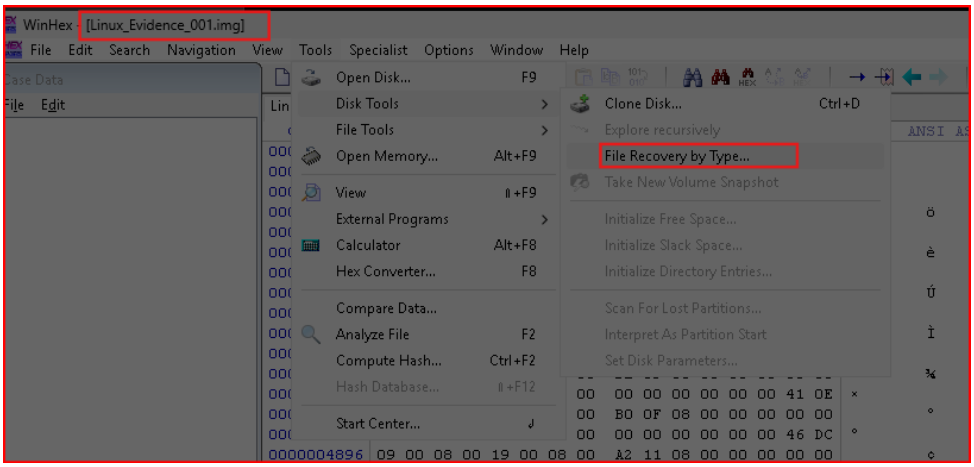
The result needs careful reading. WinHex's dialog reported **"76 file headers were found. 58 files were retrieved."** Yet the output folder contained **232** carved files (116 in the root and 116 in an ole2 subfolder), plus the log. The difference between 58 and 232 is unexplained, and no cause is asserted. The **200 KB** evaluation limit may play a part, but this was not tested (Low).

To understand what had been produced, a PowerShell script read the first eight bytes, the size and the MD5 of every file and wrote them to carved_inventory.csv. Of 233 records**, 232 begin** with **the OLE2 signature** D0 CF 11 E0 A1 B1 1A E1. The remaining record starts with the **ASCII text 10/04/20,** which fits the WinHex log file that sits in the same folder. A matching header shows only that a file begins correctly, not that it is a valid document.

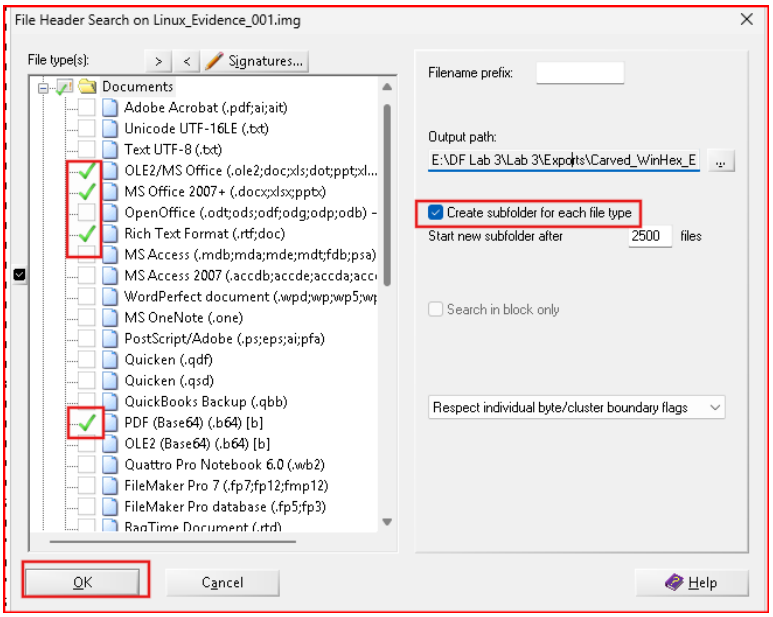
WinHex with Linux_Evidence_001.img open, read-only mode



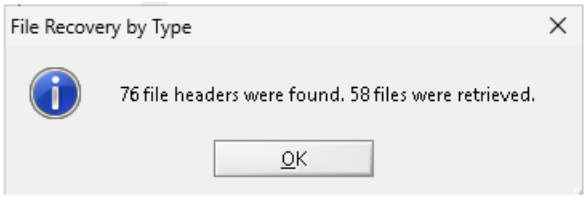
WinHex File Recovery by Type



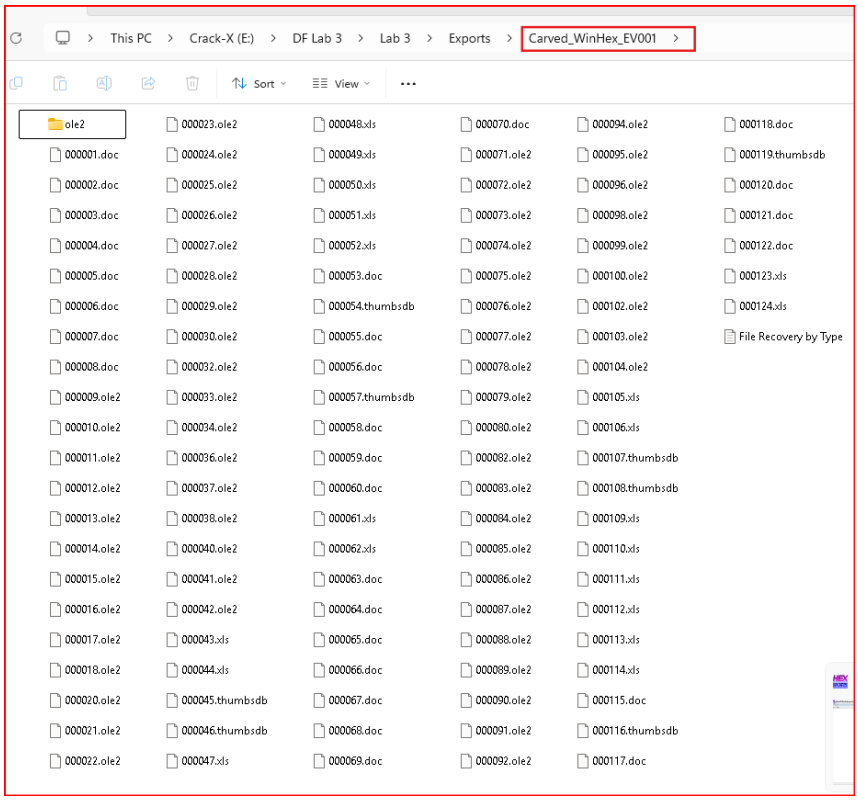
Select Target Folder dialog

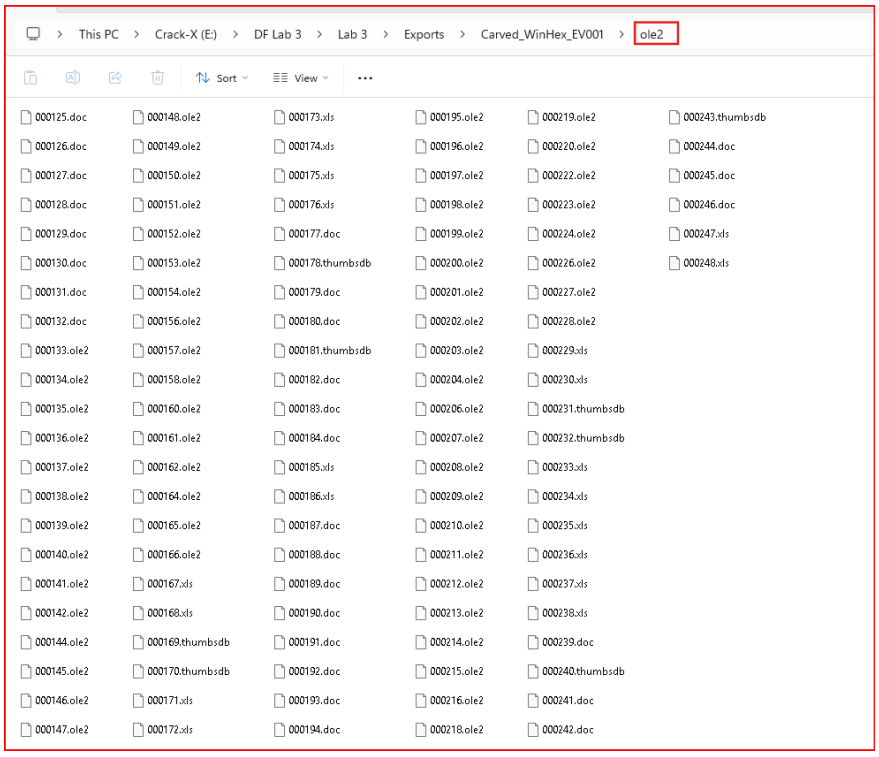


file headers found.



Carved_WinHex_EV001 folder and its ole2 subfolder





PowerShell inventory script, record count 233, grouping by first 8 bytes

```
PS E:\DF Lab 3\Exports> Get-ChildItem "E:\DF Lab 3\Lab 3\Exports\Carved_WinHex_EV001" -Recurse -File | ForEach-Object {
>>   $fs = [IO.File]::OpenRead($_.FullName)
>>   $b = New-Object byte[] 8
>>   [void]$fs.Read($b, 0, 8)
>>   $fs.Close()
>>   [pscustomobject]@{
>>     Name   = $_.Name
>>     Bytes  = $_.Length
>>     First8 = (($b | ForEach-Object { $_.ToString('X2') }) -join ' ')
>>     MD5    = (Get-FileHash $_.FullName -Algorithm MD5).Hash
>>   }
>> } | Export-Csv "E:\DF Lab 3\Lab 3\Exports\carved_inventory.csv" -NoTypeInformation
PS E:\DF Lab 3\Exports> (Import-Csv "E:\DF Lab 3\Lab 3\Exports\carved_inventory.csv").Count
233
PS E:\DF Lab 3\Exports> Import-Csv "E:\DF Lab 3\Lab 3\Exports\carved_inventory.csv" | Group-Object First8 | Select-Objec
t Count, Name

Count Name
----- ----
    1 31 30 2F 30 34 2F 32 30
  232 D0 CF 11 E0 A1 B1 1A E1

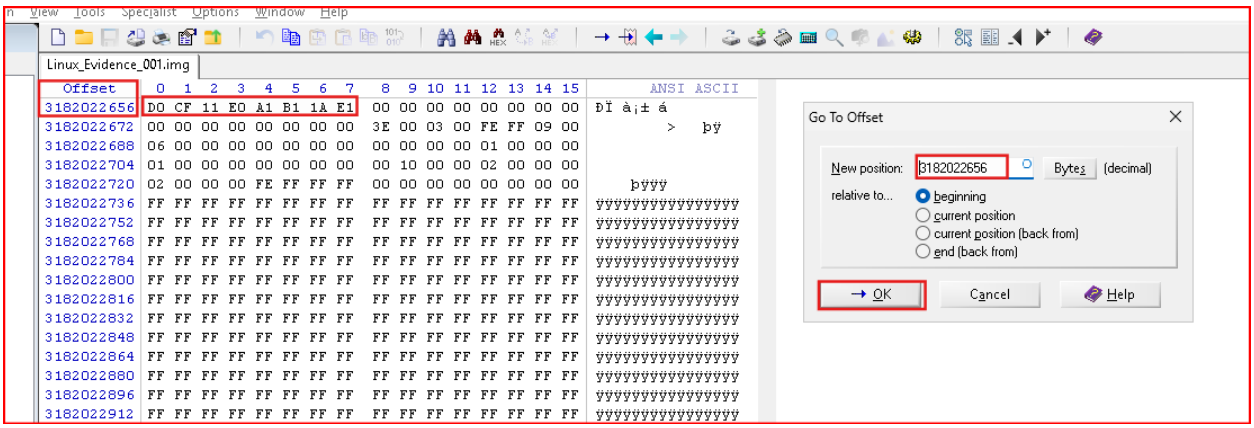

PS E:\DF Lab 3\Exports>
```

Finding the file of interest in the carved output

The next question was whether the carve had captured HideMarkedCols.xls. Phase 1 gave its first block (776,861), which is byte offset 776,861 × 4,096 = 3,182,022,656. In WinHex, *Go To Offset* at that position shows the OLE2 header D0 CF 11 E0, exactly where the file system said the file starts. A header search for the same signature produced a hit list in Position Manager.

In the inventory, **eight** carved files are 49,664 bytes with the MD5 **51AE5C0F8442964147F11C04676A411E: 000049.xls, 000052.xls, 000111.xls, 000114.xls, 000173.xls, 000176.xls, 000235.xls and 000238.xls**. That is the same MD5 size recorded for the file in phase 1. A tool that knew nothing about the file system therefore recovered the same content that the file system pointed to (High confidence). Why eight identical copies exist was not examined.

WinHex Go to Offset 3182022656 showing the D0 CF 11 E0 header

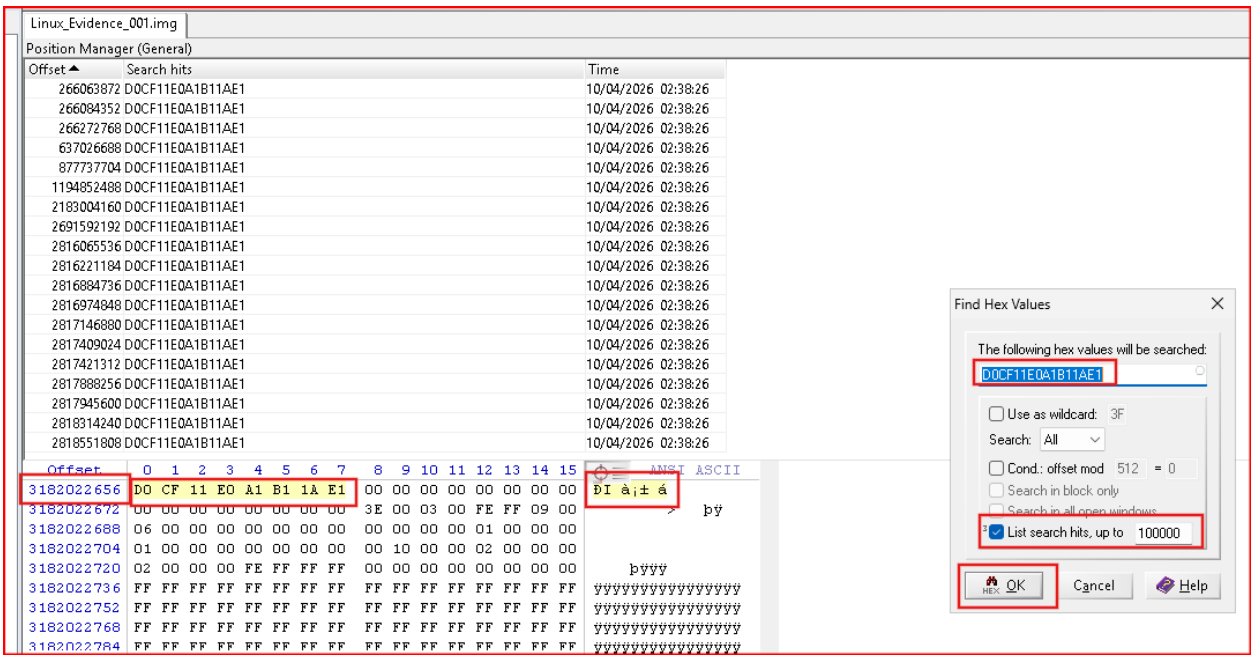


PowerShell filter listing the 8 carved items with MD5 51AE5C0F…

```
PS E:\DF Lab 3\Exports> Import-Csv "E:\DF Lab 3\Lab 3\Exports\carved_inventory.csv" | Where-Object { $_.MD5 -eq "51AE5C0
F8442964147F11C04676A411E" -or $_.Bytes -eq 49664 }

Name       Bytes First8                  MD5
----       ----- ------                  ---
000049.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000052.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000111.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000114.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000173.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000176.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000235.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
000238.xls 49664 D0 CF 11 E0 A1 B1 1A E1 51AE5C0F8442964147F11C04676A411E
```

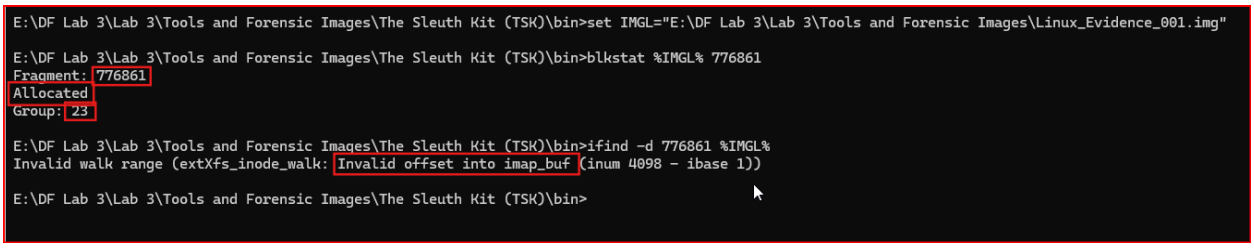
WinHex Position Manager list of search hits



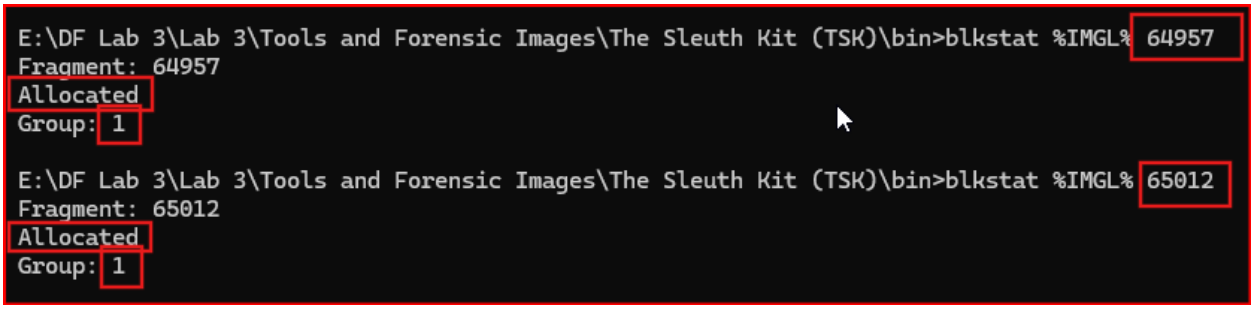
Are the carved locations in use?

blkstat reports whether a block is currently allocated. Block 776,861 is allocated (group 23), as are two further sampled blocks, 64,957 and 65,012 (both group 1). Block 64,957 corresponds to byte offset 266,063,872, the offset of the first carved file in the WinHex log. All three samples are allocated, which says nothing about the other carved items. An attempt to find the inode owning block 776,861 with ifind -d failed with "Invalid walk range", which is probably a side effect of the truncated image (Low).

blkstat 776861 and ifind -d 776861

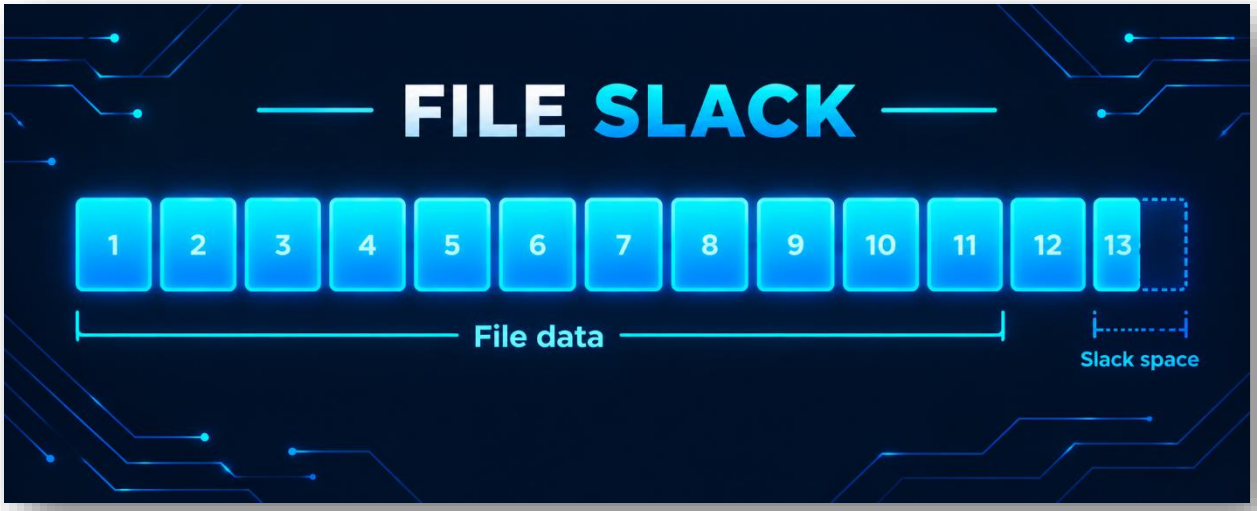


blkstat 64957 and 65012



Looking in the file slack

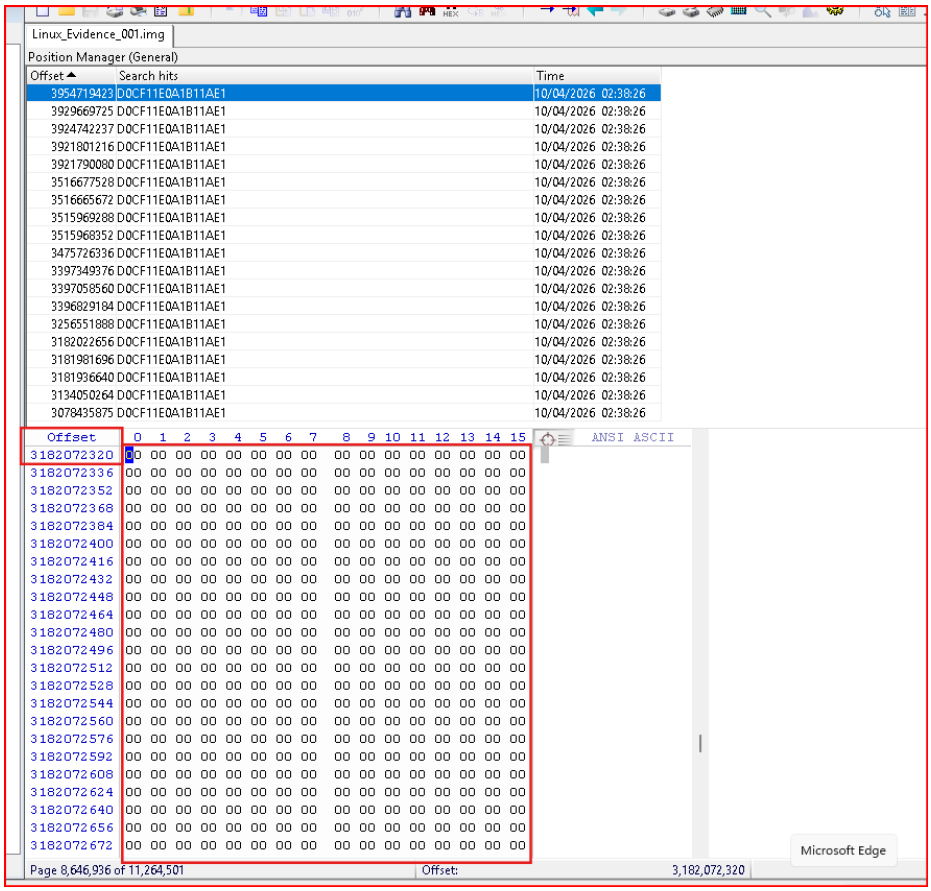
File slack is the unused part of the last block a file occupies, and it can hold remnants of whatever was stored there before. For **HideMarkedCols.xls, 3,584 bytes of slack sit** between the end of the file and the end of its **13th block**, starting at **offset 3,182,072,320.** The slack was inspected in the WinHex hex view and again by reading the same bytes with PowerShell. Both showed **3,584 bytes** and **zero non-zero bytes**.



The slack was extracted to slack_HideMarkedCols.bin and hashed (SHA-256 6CF1B57D…C16A10, MD5 B4202F7FE985B9648B4676E6F70832BD), and the MD5 of the file equals the MD5 of the same bytes held in memory.

Boundary checks closed the argument. The last 16 bytes of the file, and the first and last 16 bytes of the slack, are all 00. The first 16 bytes after the slack are <!DOCTYPE HTML P, which belong to a different, unrelated data unit that was not analysed. The slack held only null bytes and no residual data was observed (High).

WinHex view of the slack region from offset 3182072320, all 00



PowerShell output, bytes read 3584, non-zero bytes 0

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> $fs.Close()
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> "Bytes read       : $n"
Bytes read       : 3584
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> "Non-zero bytes   : " + (($b | Where-Object { $_ -ne 0 }).Count)
Non-zero bytes   : 0
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> "Zero bytes       : " + (($b | Where-Object { $_ -eq 0 }).Count)
```

slack extract creation with SHA-256 and MD5

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> mkdir (Split-Path $out) -Force | Out-Null
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> [IO.File]::WriteAllBytes($out,$b)
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> (Get-Item $out).Length
3584
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-FileHash $out -Algorithm SHA256

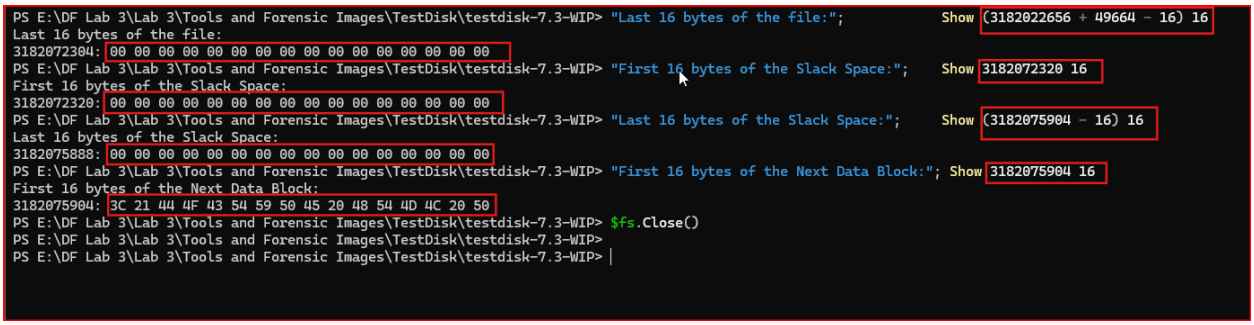
Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          6CF1B57D59E7111BC218DFB01DDA93AC0F776715599A1C69F89035BD20C16A10       E:\DF Lab 3\Lab 3\Exports\Lab3\slack_HideMarkedCols.bin


PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-FileHash $out -Algorithm MD5

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
MD5             B4202F7FE985B9648B4676E6F70832BD                                       E:\DF Lab 3\Lab 3\Exports\Lab3\slack_HideMarkedCols.bin


PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> $z=New-Object byte[] 3584
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> $ms=[IO.MemoryStream]::new($z); (Get-FileHash -InputStream $ms -Algorithm MD5).Hash
B4202F7FE985B9648B4676E6F70832BD
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP>
```

boundary checks at end of file, end of slack and start of next block

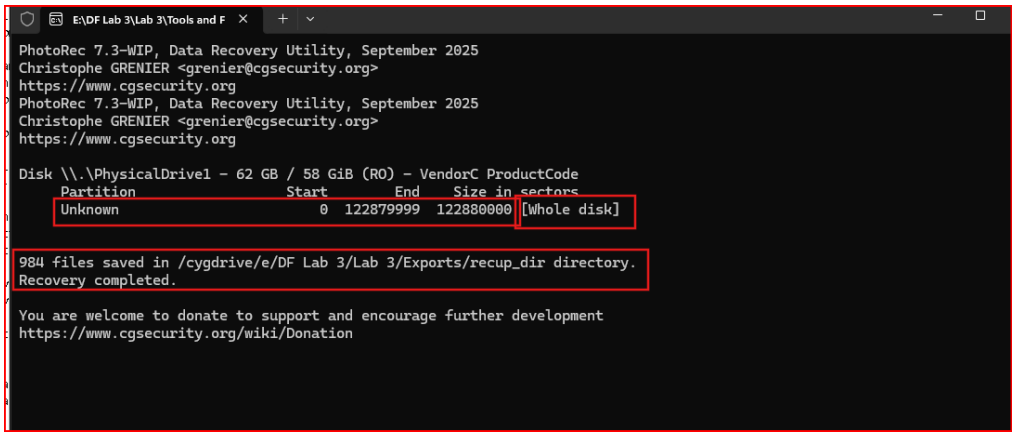


PhotoRec as second carving tool

PhotoRec was chosen as an independent tool. The first attempt went wrong: PhotoRec was pointed at the examiner's **physical drives** (PhysicalDrive0 and PhysicalDrive1) instead of the image. It produced a "Photo Rec" folder and four recup_dir folders, and one screen reported 984 files saved.

Note: Those outputs did not come from EV-001, and EV-001 was not involved. The folders were renamed with the prefix INVALID_wrong_source_ and kept, not deleted, and they are excluded from every finding.

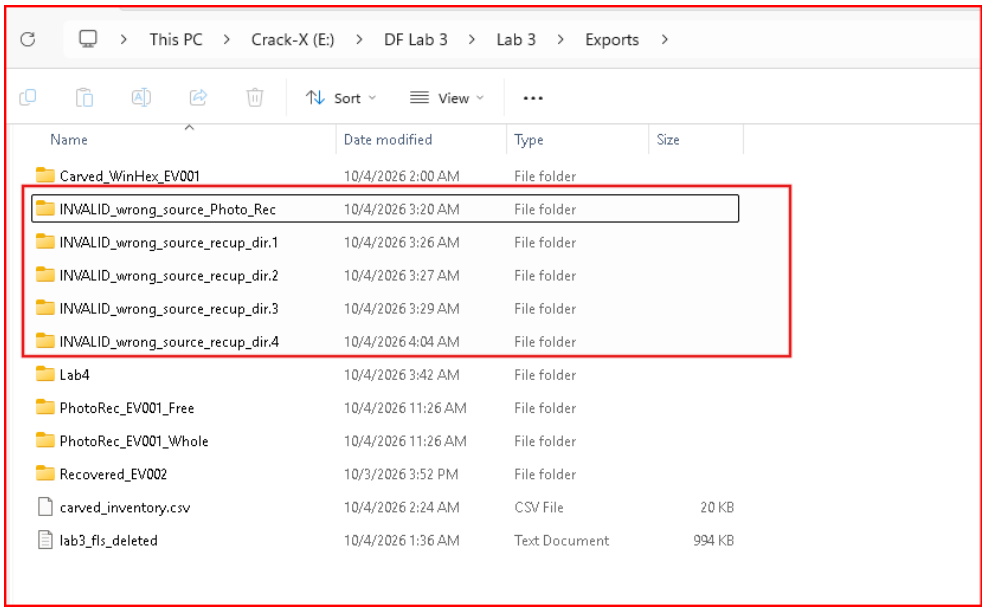
PhotoRec drive listing showing the physical drives



PhotoRec screens of the first attempt and "984 files saved"



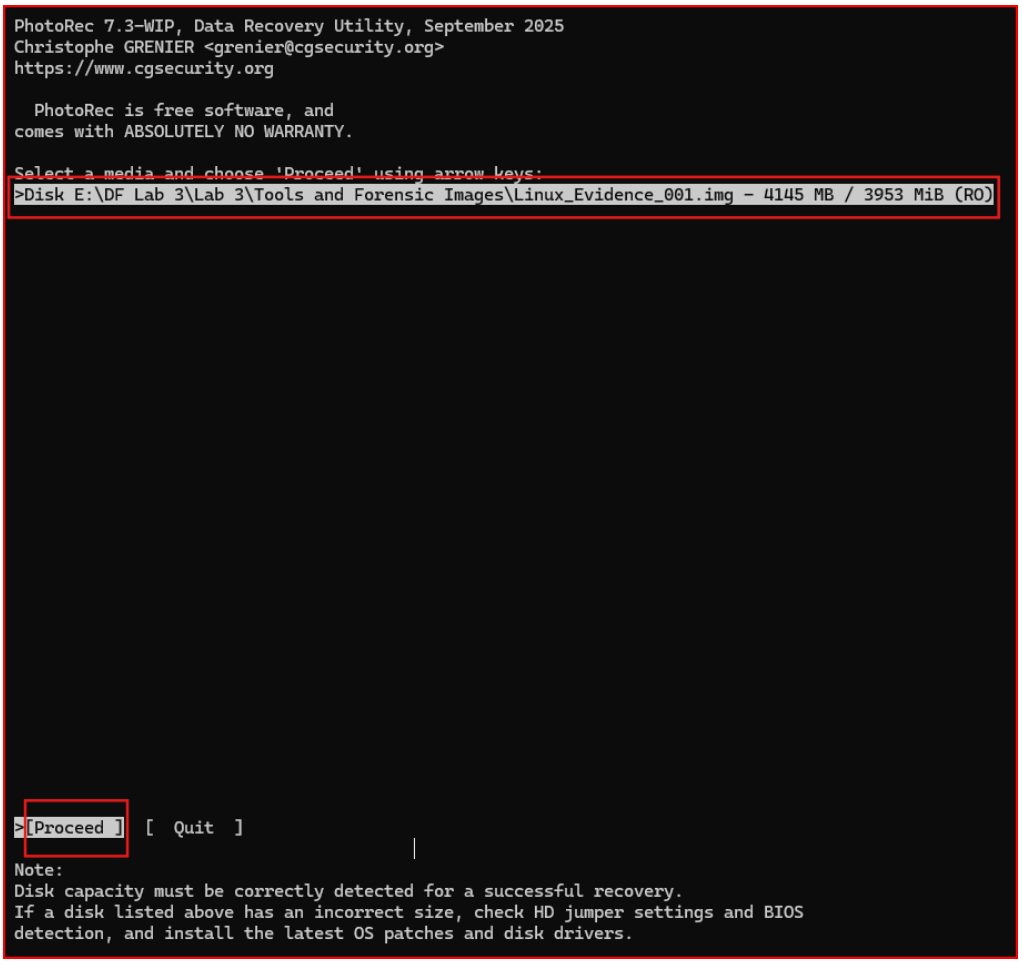
PowerShell rename and Explorer listing of INVALID_wrong_source_ folders



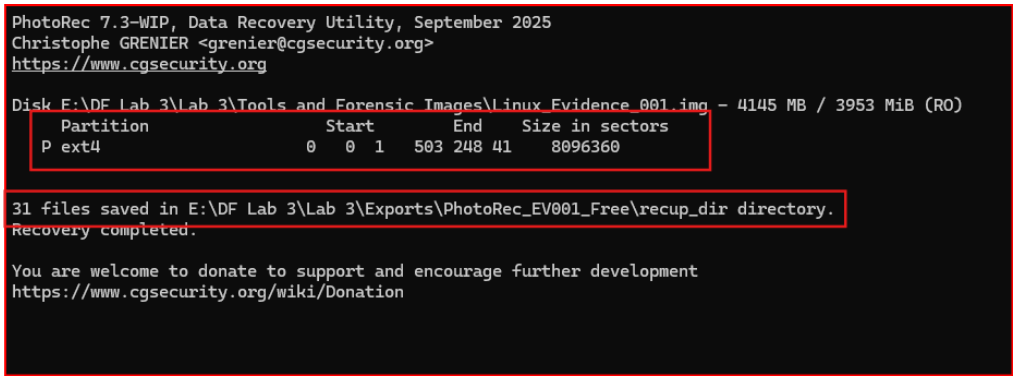
PhotoRec was then run correctly on the image itself, once in free space only and once on the whole disk, with the image re-hashed before and after. For the free space run the source was Linux_Evidence_001.img (read-only), the layout was shown as ext4, and the options were Paranoid and Keep corrupted files set to Yes, Expert mode and Low memory set to No, with the doc file type selected. Both runs reported **31 files saved**, and the image hash was unchanged after each.

The free-space output holds **19 .doc, 10 .xls, 2 .db and 1 .xml file.** The 31 doc, xls and db files have only **15 distinct MD5 values**, showing **many duplicates**. One **49,664-byte file, f6214888.xls, has the MD5 51AE5C0F8442964147F11C04676A411E**, equal to the control file, and its name encodes **sector 6,214,888,** which is the control file's start sector. Three PhotoRec output names carry descriptive text; they were neither interpreted nor opened.

PhotoRec Options and File Opt screens for the EV-001 run



PhotoRec ext2/ext3/ext4 and Free scan selection, "31 files saved"



extension counts, file listing, distinct MD5 count and the 49,664-byte file hash

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-ChildItem $A -Recurse -File | Where-Object Name -ne "photorec.log" | Group-Object Extension | Select Count,Name

Count Name
----- ----
    1 .xml
   19 .doc
   10 .xls
    2 .db
```

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-ChildItem $A -Recurse -File | Select-Object Name,Length,@{n='Dir';e={$_.Directory.Name}} | Sort Name | Format-Table -AutoSize

Name                                                                                Length Dir
----                                                                                ------ ---
f0519656_Wednesday_May_11_2005.doc                                                   20480 recup_dir.1
f0519696_My_swiss_bank_account_number_is_43324324324_432432432432432_2343244423.doc  18944 recup_dir.1
f0520064.doc                                                                         24064 recup_dir.1
f4263680_CURRICULUM_VITAE.doc                                                        88576 recup_dir.1
f5257016.doc                                                                          1536 recup_dir.1
f5500128.doc                                                                          1536 recup_dir.1
f5500432.doc                                                                          1536 recup_dir.1
f5501728.xls                                                                          1536 recup_dir.1
f5501904.xls                                                                          1536 recup_dir.1
f5502240.xls                                                                          1536 recup_dir.1
f5502752.xls                                                                          1536 recup_dir.1
f5502776.doc                                                                          1536 recup_dir.1
f5503688.doc                                                                          1536 recup_dir.1
f5503800.doc                                                                        344064 recup_dir.1
f5504520.doc                                                                          2048 recup_dir.1
f5504984.doc                                                                          1536 recup_dir.1
f5505000.doc                                                                          1536 recup_dir.1
f5505680.doc                                                                        329216 recup_dir.1
f5506520.xls                                                                          1536 recup_dir.1
f5506688.doc                                                                          1536 recup_dir.1
f5508096.doc                                                                          1536 recup_dir.1
f5858688.xls                                                                         30208 recup_dir.1
f5908800.db                                                                          17920 recup_dir.1
f6214720.xls                                                                         41472 recup_dir.1
f6214808.xls                                                                         38400 recup_dir.1
f6214888.xls                                                                         49664 recup_dir.1
f6634432.doc                                                                         24064 recup_dir.1
f6634880.db                                                                          41472 recup_dir.1
f6635448_Wednesday_May_11_2005.doc                                                   20480 recup_dir.1
f6788528_My_swiss_bank_account_number_is_43324324324_432432432432432_2343244423.doc  18944 recup_dir.1
f7659768.xls                                                                        102912 recup_dir.1
report.xml                                                                            7814 recup_dir.1
```

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-ChildItem $A -Recurse -File | Where-Object Extension -in '.doc','.xls','.db' | Get-FileHash -Algorithm MD5 | Group-Object Hash | Measure-Object | Select Count

Count
-----
   15
```

```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images\TestDisk\testdisk-7.3-WIP> Get-ChildItem $A -Recurse -File | Where-Object { $_.Length -eq 49664 } | Get-FileHash -Algorithm MD5 | Select Hash,Path

Hash                             Path
----                             ----
51AE5C0F84442964147F11C04676A411E E:\DF Lab 3\Lab 3\Exports\PhotoRec_EV001_Free\recup_dir.1\f6214888.xls
```

whole-disk run, "31 files saved"

```
PhotoRec 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk E:\DF Lab 3\Lab 3\Tools and Forensic Images\Linux_Evidence_001.img - 4145 MB / 3953 MiB (RO)
     Partition                  Start        End    Size in sectors
     Unknown                  0   0  1  503 248 41    8096360 [Whole disk]

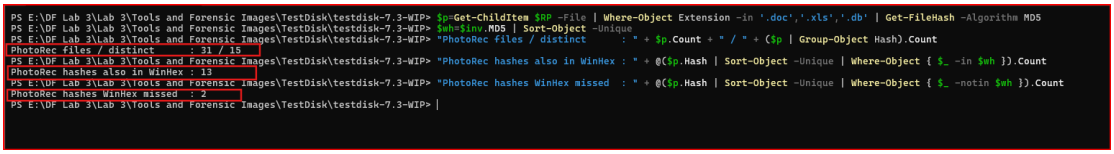

31 files saved in E:\DF Lab 3\Lab 3\Exports\PhotoRec_EV001_Whole\recup_dir directory.
Recovery completed.

You are welcome to donate to support and encourage further development
https://www.cgsecurity.org/wiki/Donation
```

Comparing the two carving tools

The **31 PhotoRec** files were compared against **WinHex's 232** files by MD5. PhotoRec produced **15** distinct hashes and WinHex **13** across its larger set. Thirteen PhotoRec hashes also occur in the WinHex set, and **two do not**: **f5503800.doc and f5505680.doc.** The reasons for the duplicates and the content of those two files were not examined. The figures are reported as observed.

PowerShell comparison of PhotoRec and WinHex hashes (31 / 15, 13, 2)

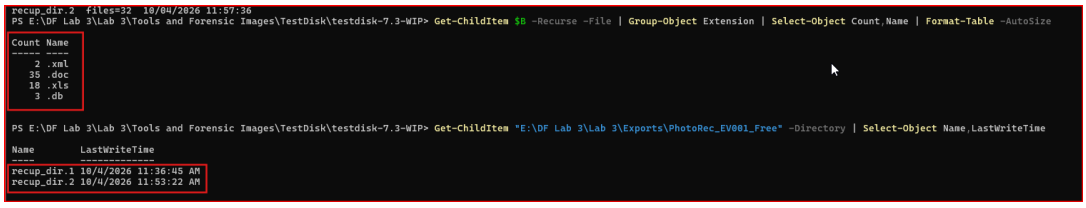


the two PhotoRec hashes absent from the WinHex set

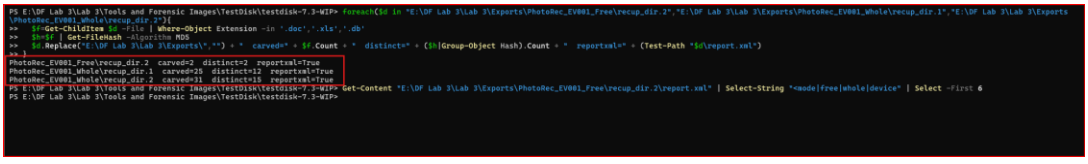


The PhotoRec work folders were also counted in last phase, because the earlier totals were not reconciled. The recursive count of the whole-disk output was 60 items, and a separate extension count reads 2 .xml, 35 .doc, 18 .xls and 3 .db. Per-folder counts were: Free recup_dir.2, 2 carved files (2 distinct); Whole recup_dir.1, 25 (12 distinct); Whole recup_dir.2, 31 (15 distinct), each with a report.xml. The figures are arithmetically consistent: 25 + 31 = 56 carved files, plus 2 report files makes 58, plus 2 folders makes 60. Which output the extension count refers to is not visible in the screenshot, so this reconciliation stays at Moderate confidence, and the idea that each destination holds two separate sessions is an inference (Low).

Exports folder listing with last-write times and the Whole item count of 60



Extension counts and per-folder carved, distinct and report.xml values



Building Timelines

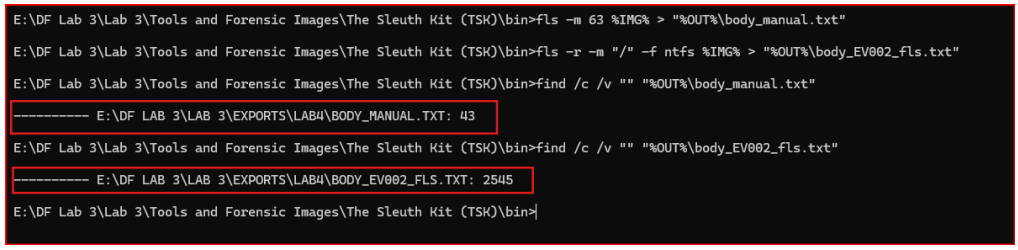
A timeline places every file-system stamp into one ordered list, which helps spot activity clusters and test earlier conclusions. A Sleuth Kit **body file** is written first (fls for named entries, ils for nameless inodes) and mactime then converts it into a timeline.

Reproducing and correcting the manual's command

The command, fls -r -m "/" -f ntfs <image>, produced **2,545 lines**. The version adds recursion (-r), a mount-point prefix (/) and an explicit file-system type (-f ntfs). The jump is consistent with recursion being the main factor (Moderate), but the effect of each option was not tested separately. A point worth checking on your build with fls -h: in fls, -m sets the mount-point prefix written into the body file, while the sector offset is -o. So the manual's 63 was most likely read as a path prefix and not as an offset. This also fits Lab 2, where EV-002 was shown to need no offset at all.

Note: ils -r -m was also run to capture deleted inodes that have no name, but its output was not merged into the timelines, so nameless deleted inodes are not covered here.

fls manual and corrected commands with line count 43 and 2545



ils -r -m command writing body_EV002_ils.txt



Producing the EV-002 timelines

Four timelines were generated from the corrected body file, all in UTC with ISO dates (-y -z UTC): a plain text timeline, a delimited CSV (-d), a daily index (-i day), and a slice covering 2019-12-19 to 2019-12-20, the busiest period. The body file (MD5 f369c33f1d2b1739b29f44b1b50d1cce) and the text timeline (MD5 715b95a2ca8dbe4030646fa57dd2a70f) were hashed with certutil. Hashing derived files means their integrity can be checked later.

Four mactime commands (text, CSV, day index, date slice)

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>perl mactime.pl -b "%OUT%\body_EV002_fls.txt" -y -z
UTC > "%OUT%\timeline_EV002_UTC.txt"

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>perl mactime.pl -b "%OUT%\body_EV002_fls.txt" -y -d
-z UTC > "%OUT%\timeline_EV002_UTC.csv"

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>perl mactime.pl -b "%OUT%\body_EV002_fls.txt" -y -z
UTC -i day "%OUT%\index_day.txt" > nul

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>perl mactime.pl -b "%OUT%\body_EV002_fls.txt" -y -z
UTC 2019-12-19..2019-12-20 > "%OUT%\timeline_20191219.txt"

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>
```

certutil MD5 of body_EV002_fls.txt and timeline_EV002_UTC.txt

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>certutil -hashfile "%OUT%\body_EV002_fls.txt" MD5
MD5 hash of E:\DF Lab 3\Lab 3\Exports\Lab4\body_EV002_fls.txt:
f369c33f1d2b1739b29f44b1b50d1cce
CertUtil: -hashfile command completed successfully.

E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>certutil -hashfile "%OUT%\timeline_EV002_UTC.txt" MD5
MD5 hash of E:\DF Lab 3\Lab 3\Exports\Lab4\timeline_EV002_UTC.txt:
715b95a2ca8dbe4030646fa57dd2a70f
CertUtil: -hashfile command completed successfully.
```

What the EV-002 timeline shows

Searching the timeline for the second 09:55:24 returned two rows. One is the $MFT entry on 2019-12-19 with all four time types equal (flags macb), which matches the volume-creation instant measured. The other is MicrosoftNTP.cap, a capture file in a Wireshark sample folder, whose modification time shares the same clock second on **5 April 2011**. The same second on two dates is likely coincidence, and no conclusion is drawn about the capture file. These values were read from a small screenshot and should be verified.

findstr output for 09:55:24

```
E:\DF Lab 3\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>findstr /c:"09:55:24" "E:\DF Lab 3\Lab 3\Exports\Lab4\timeline_EV002_UTC.txt"
2011-04-05T09:55:24Z   434397 m... r/rrwxrwxrwx 0       0       1009-128-1 /Other Files/Wireshark Sample Capture Files/Specific Protocols and Protocol Families/Network Time Protocol/MicrosoftNTP.cap
2019-12-19T09:55:24Z   1310720 macb r/rr-xr-xr-x 0      0       0-128-6  /$MFT
```

Searching the CSV for ,96-128- returned **six rows for the deleted Word_Doc1.docx and its Zone.Identifier stream.** Each time appears for both the data stream and the Zone.Identifier stream. They show the **content-modified time (2016-06-16 06:32:26Z)** on lines **1013–1014**, the **MFT-changed time (2019-12-19 09:10:51Z**) on **lines 2264–2265**, and the accessed and **created times (2019-12-19 09:55:49Z**) on **lines 3084–3085**. These agree exactly with the istat values from phase 2.

Note: The change time falls 44 minutes 58 seconds before creation, which remains unexplained. The timeline cannot show whether the file was deleted before or after these times.

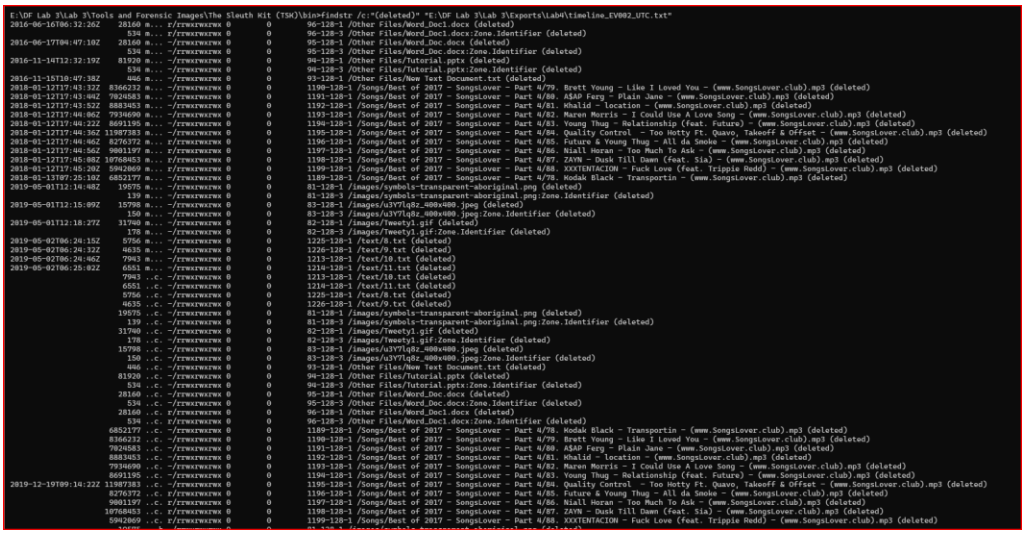
Select-String output for ,96-128- in the timeline CSV

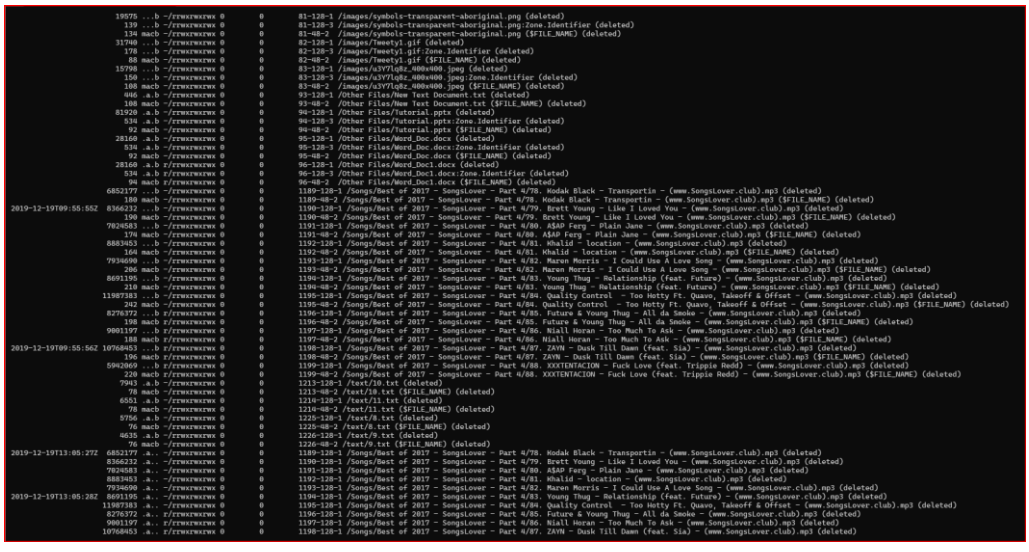
```
PS E:\DF Lab 3\Lab 3\Tools and Forensic Images> Select-String -Path "E:\DF Lab 3\Lab 3\Exports\Lab4\timeline_EV002_UTC.csv" -Pattern ",96-128-"

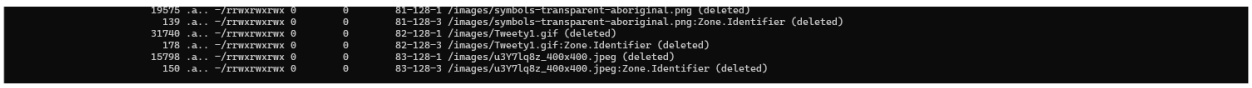
E:\DF Lab 3\Lab
3\Exports\Lab4\timeline_EV002_UTC.csv:1013:2016-06-16T06:32:26Z,28160,m...,r/rrwxrwxrwx,0,0,96-128-1,"/Other
Files/Word_Doc1.docx (deleted)"
E:\DF Lab 3\Lab
3\Exports\Lab4\timeline_EV002_UTC.csv:1014:2016-06-16T06:32:26Z,534,m...,r/rrwxrwxrwx,0,0,96-128-3,"/Other
Files/Word_Doc1.docx:Zone.Identifier (deleted)"
E:\DF Lab 3\Lab
3\Exports\Lab4\timeline_EV002_UTC.csv:2264:2019-12-19T09:10:51Z,28160,..c.,r/rrwxrwxrwx,0,0,96-128-1,"/Other
Files/Word_Doc1.docx (deleted)"
E:\DF Lab 3\Lab
3\Exports\Lab4\timeline_EV002_UTC.csv:2265:2019-12-19T09:10:51Z,534,..c.,r/rrwxrwxrwx,0,0,96-128-3,"/Other
Files/Word_Doc1.docx:Zone.Identifier (deleted)"
E:\DF Lab 3\Lab
3\Exports\Lab4\timeline_EV002_UTC.csv:3084:2019-12-19T09:55:49Z,28160,.a.b,r/rrwxrwxrwx,0,0,96-128-1,"/Other
Files/Word_Doc1.docx (deleted)"
E:\DF Lab 3\Lab
3\Exports\Lab4\timeline_EV002_UTC.csv:3085:2019-12-19T09:55:49Z,534,.a.b,r/rrwxrwxrwx,0,0,96-128-3,"/Other
Files/Word_Doc1.docx:Zone.Identifier (deleted)"
```

A search for the label (deleted) showed many entries under /Other Files, /images, /text and /Songs, including $FILE_NAME attribute rows for some deleted names. Their content was not opened and no conclusion is drawn from the names. They are recorded as leads.

findstr (deleted) listing



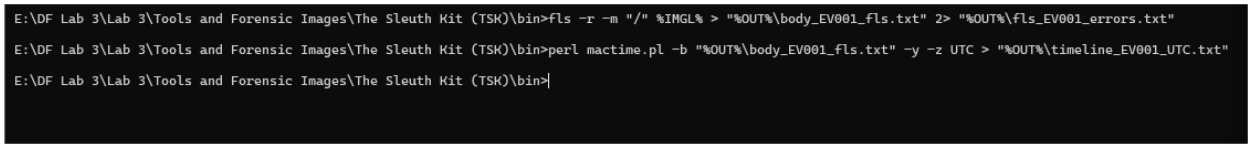




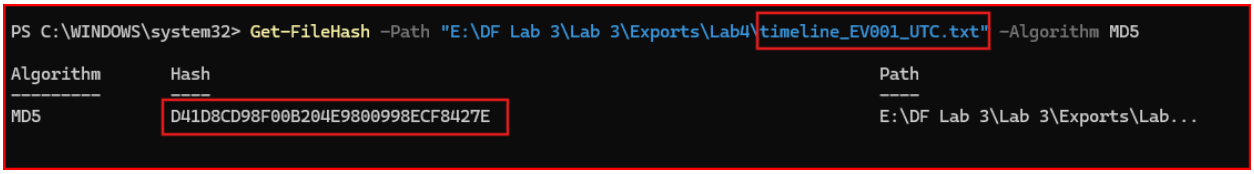
A failed timeline and a failed offset test

The first attempt to timeline EV-001 produced a file whose MD5 is **D41D8CD98F00B204E9800998ECF8427E.** That is the well-known **MD5 of zero-length input,** so the file was almost certainly empty. No error appeared on screen, and the cause was not captured, so the file was discarded and not used as evidence.

first EV-001 fls and mactime commands with no output



Get-FileHash MD5 of the empty timeline_EV001_UTC.txt



A related test showed why offsets matter. Running fls -r -m "/" on a Windows .dd image without an offset returned "**Cannot determine file system type**". The filename in the screenshot reads Windows_Evidence_002.dd, which is EV-003 and matches phase 2: its first file system starts at **sector 2,048,** so **nothing exists at sector 0**. A second run with -o 2048 was started to produce a partition-level timeline, but its results are not part of the findings.

fls failing with "Cannot determine file system type"



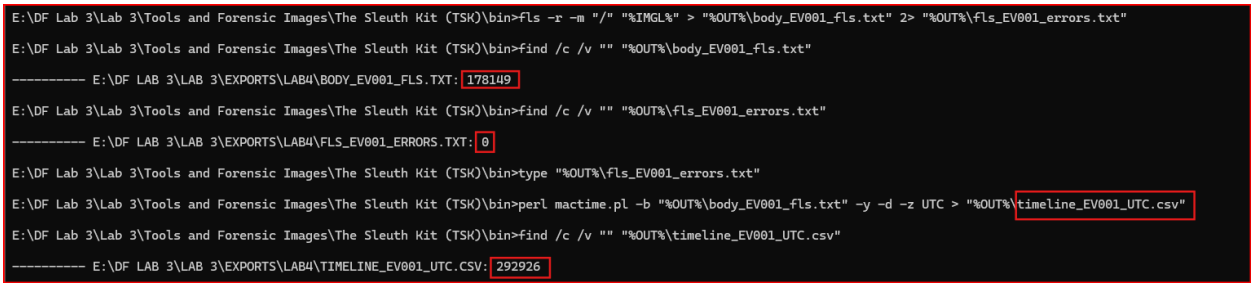
fls with -o 2048 and the mactime commands



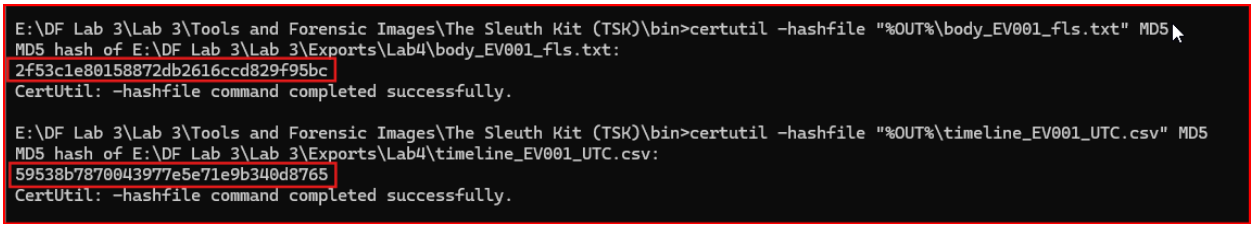
Rebuilding the EV-001 timeline

The EV-001 timeline was rebuilt carefully. fls was re-run with its error output sent to a separate file. The body file has **178,149 lines** and the error file has **0 lines**, so the run was clean. mactime then produced a CSV timeline of **292,926 rows**. There are more rows than body lines because mactime writes one row for each distinct time of each entry. The body file (MD5 2F53C1E80158872DB2616CCD829F95BC) and the timeline (MD5 59538B7870043977E5E71E9B340D8765) were hashed with both certutil and Get-FileHash, and the values agree.

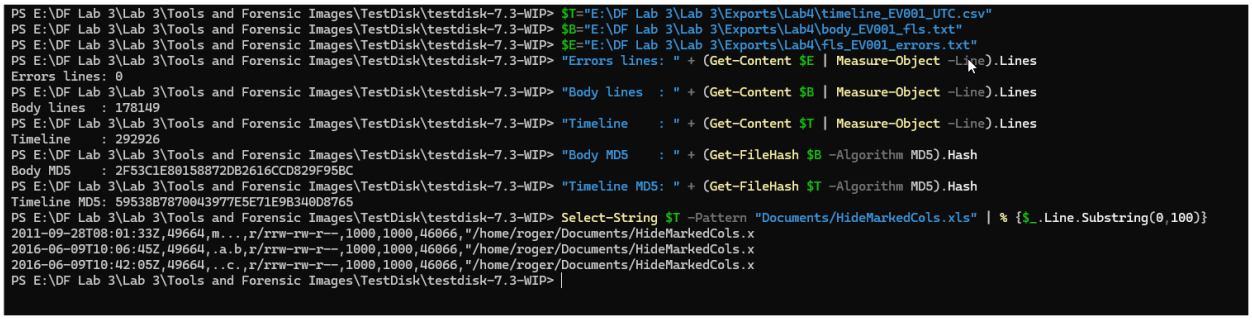
rebuilt body file and CSV timeline



certutil MD5 of body_EV001_fls.txt and timeline_EV001_UTC.csv]



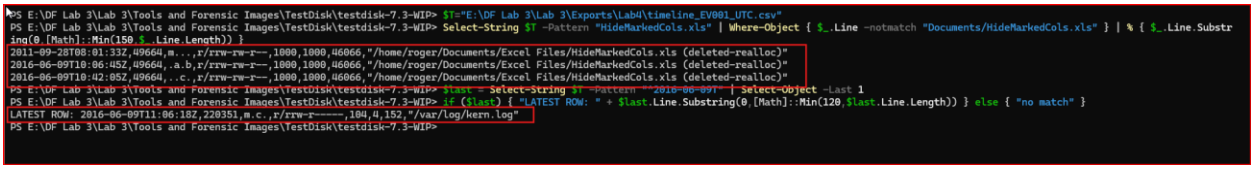
check of error lines, body lines, timeline lines and MD5



Using the timeline to test phase 1

HideMarkedCols.xls. The timeline shows three rows for inode 46066: modified 2011-09-28 08:01:33Z, created and accessed 2016-06-09 10:06:45Z, and changed 10:42:05Z. These are Lab 1's times converted to UTC. The same inode also appears under a second, deleted name, /home/roger/Documents/Excel Files/HideMarkedCols.xls, labelled *(deleted-realloc)*. As the inode is allocated, that deleted name is a stale directory entry that points to the same inode.

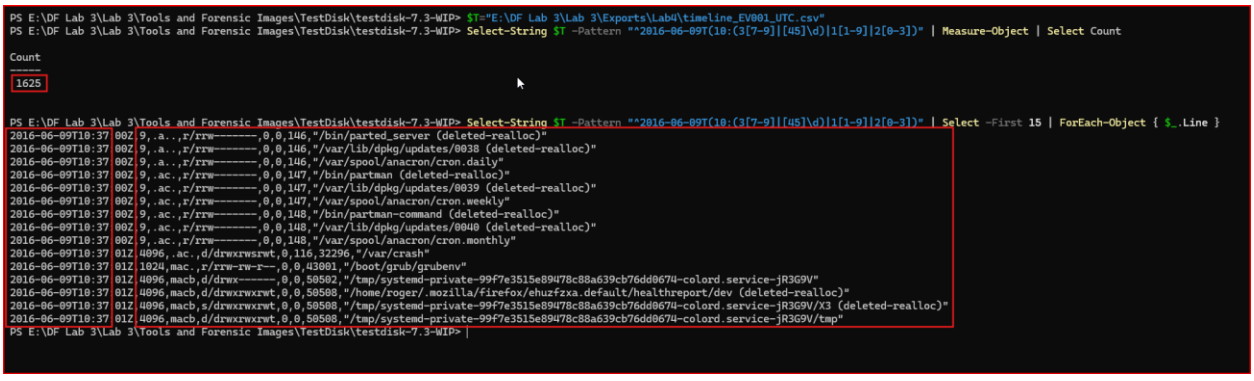
timeline rows for HideMarkedCols.xls under /home/roger/Documents/



The last mount. Last mount time **(15:36:52 UZT) is 10:36:52Z**. The timeline contains **1,626** rows from **10:37Z** onward on **9 June 2016**.

The first rows are access times on /var/spool/anacron entries at 10:37:00Z, followed by directory times on /var/crash, /boot/grub/grubenv and a systemd-private directory under /tmp. The latest row of that day is /var/log/kern.log (inode 152) at 11:06:18Z, 29 minutes 26 seconds after the last mount. Lab 1 had recorded activity up to 15:45:57 UZT. This timeline extends that by about 20 minutes to 16:06:18 UZT, which supports Lab 1's reading that the system kept running after the recorded last mount. The stop time is still not fixed.

Select-String count (1,626) and the first 15 matching rows



Verifying What Was Recovered

Recovery produces files, but a file's name proves nothing about its contents. Lab 5 tested whether the 22 files recovered from EV-002 match their extensions.

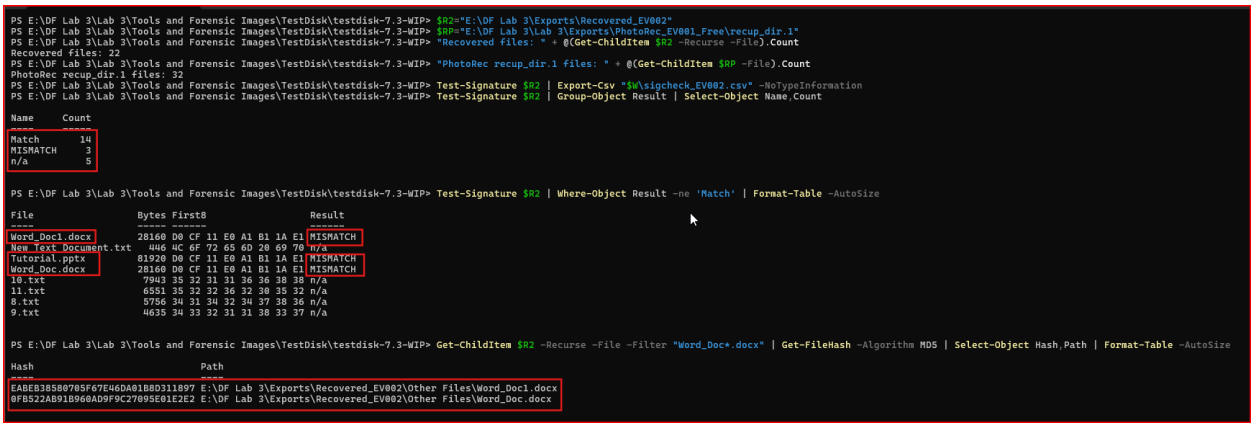
Signature test

A PowerShell function, Test-Signature, compared each file's first bytes with the signature expected for its extension. Of 22 files, 14 returned Match, 3 returned MISMATCH and 5 returned n/a. By elimination against the Lab 2 inventory, the 14 matches are the three image files and the eleven audio files [VERIFY from sigcheck_EV002.csv, since the screenshot lists only the non-matching rows]. The five n/a results are text files (New Text Document.txt, 8.txt, 9.txt, 10.txt and 11.txt), which have no defined signature. The function's definition and signature table were not captured, so this reading of "n/a" is an inference (Low).

File	Bytes	First 8 bytes	Result
Word_Doc1.docx	28,160	D0 CF 11 E0 A1 B1 1A E1	MISMATCH
Word_Doc.docx	28,160	D0 CF 11 E0 A1 B1 1A E1	MISMATCH
Tutorial.pptx	81,920	D0 CF 11 E0 A1 B1 1A E1	MISMATCH

Files with the extensions .docx and .pptx are modern Office Open XML documents, which normally begin with a ZIP header. These three begin instead with the OLE2 header used by legacy Office files. File sizes match the entries in the Lab 4 timeline.

Test-Signature commands, summary (14 / 3 / 5) and the list of non-matching files

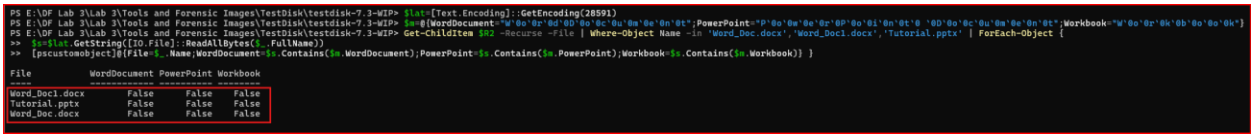


What is inside the mismatched files?

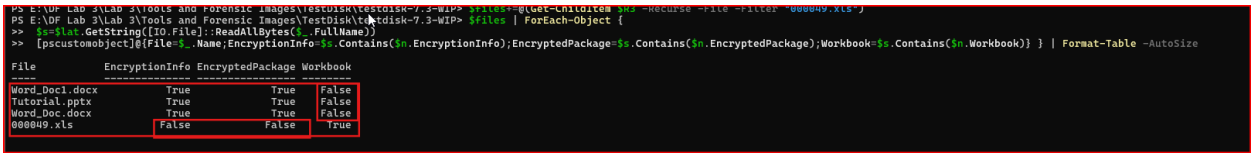
To explain the mismatch without opening the files, each was read as raw bytes (decoded as ISO-8859-1) and searched for the UTF-16 names of the streams that live inside an OLE2 container. A legacy Word, PowerPoint or Excel file contains WordDocument, PowerPoint Document or Workbook. An encrypted Office 2007+ document stored in an OLE2 wrapper contains EncryptionInfo and EncryptedPackage instead.

All three mismatched files returned **False** for the legacy stream names and **True** for both EncryptionInfo and EncryptedPackage. As a positive control, the carved 000049.xls (a known legacy Excel file) returned the opposite: Workbook found, and neither encryption name found. The control shows the method can detect a known stream. The structure is therefore consistent with encrypted-package wrappers, which would explain why the header and the extension disagree (Moderate). A string search is not a parser, so a False result is not proof of absence, and the interpretation was not confirmed with a parser. No file was opened or decrypted, and nothing is inferred about content or intent.

byte-search results for WordDocument / PowerPoint Document / Workbook



byte-search results for EncryptionInfo / EncryptedPackage / Workbook including 000049.xls

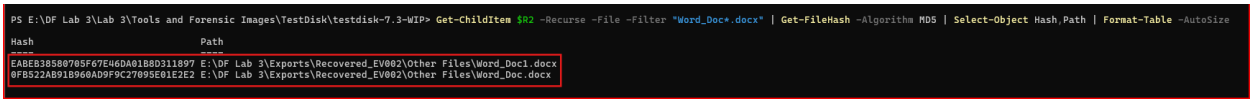


Closing the loop on inode 96

Phase 2 left one comparison open: whether the tsk_recover copy of Word_Doc1.docx matches the icat copy.

Get-FileHash on the recovered Word documents **gave Word_Doc1.docx** an MD5 of **EABEB38580705F67E46DA01B8D311897,** identical to the phase 2 icat value **eabeb38580705f67e46da01b8d311897.** Two different recovery routes produced the same content (High). The second Word document, Word_Doc.docx (inode 95), is also 28,160 bytes, but its MD5 differs (0FB522AB91B960AD9F9C27095E01E2E2), so the two files are not copies of each other. No cause is asserted.

MD5 of Word_Doc1.docx and Word_Doc.docx



One file, five layers. HideMarkedCols.xls was traced through every layer the module covers. Its name and metadata gave **inode 46066** and **blocks 776861 to 776873**. Its content identity was confirmed by one MD5 across Autopsy, an exported copy and an icat copy. Its raw location matched the block arithmetic, with the header at byte **3,182,022,656**. Two carving tools recovered identical copies: eight from WinHex and one from PhotoRec. Its slack held only null bytes. Its timeline entries reproduced the phase 1 times. All these results agree. None of this analysis concerns the file's content or purpose.

Metadata recovery against carving. On EV-002, Metadata-based recovery returned 22 files with names, paths and timestamps. On EV-001, carving returned only content with no names or dates. WinHex and PhotoRec produced different counts (232 against 31 files; 13 against 15 distinct hashes), yet both recovered the file of interest with an identical hash.

A pattern, not an oddity. The timestamp irregularity first seen in inode 96 (Lab 2) was found to hold in all 22 recoverable entries and was reproduced in the timeline (Lab 4). Because it recurs so consistently, it is flagged as a systematic feature of this volume rather than a single anomaly.

Consolidated timeline of events recorded inside the evidence (UTC)

UTC	Event	Evidence
2011-04-05 09:55:24	MicrosoftNTP.cap modified	EV-002
2011-09-28 08:01:33	HideMarkedCols.xls content last modified	EV-001
2016-04-05 21:59:09	crontab content last modified	EV-001
2016-06-09 09:18:26	File system last checked	EV-001
2016-06-09 09:18:35 to 09:22:45	fstab, hostname, hosts and timezone written	EV-001
2016-06-09 10:06:45	HideMarkedCols.xls created on this volume	EV-001
2016-06-09 10:36:48 / 10:36:52	Superblock last written / file system last mounted	EV-001
2016-06-09 10:42:05	HideMarkedCols.xls inode changed	EV-001
2016-06-09 11:06:18	Latest 2016-06-09 row, /var/log/kern.log	EV-001
2016-06-16 06:32:26	Word_Doc1.docx content modified	EV-002
2019-12-19 09:10:51	Word_Doc1.docx MFT record modified	EV-002
2019-12-19 09:55:24	System files created; $MFT macb row	EV-002
2019-12-19 09:55:49	Word_Doc1.docx created	EV-002
2019-12-19 09:55:57	Root directory modified	EV-002
2019-12-19 13:05:52	Root directory accessed	EV-002

For display time add 5 hours (UZT). For the EV-001 system's own clock, add 5 hours 30 minutes. The times of the examiner's own actions are listed in the chain of custody (Section 14).

Limitations and Unresolved Items

These matter as much as the findings and are recorded openly.

The EV-001 image is truncated. About a fifth of the file system's blocks are absent. Other users' home directories, the failed ils and ifind -d commands, and any content in the upper block groups cannot be assessed. Missing content is not evidence of deletion.

The WinHex count is unexplained. The tool reported 58 files retrieved, but 232 files are on disk. The totals depend on which figure is used. The evaluation-version limit of 200 KB was not tested as a cause.

Several counts remain open. The PhotoRec output folders are not fully reconciled, and the 51 reference-0 lines on EV-002 were not matched by name. The effect of each option in the corrected fls command was not tested separately. The nameless-inode body file was never merged into a timeline.

The Windows timezone is unknown. No local-time column can be given for EV-002 or EV-003. Nothing beyond fsstat was run on EV-003's partitions, and its exact size is derived from mmls and not from file properties.

Interpretation limits. Timeline times are file-system metadata. Copying, extraction or tool actions can preserve or change them, so they do not show who acted or when. The stream-name search is a string match and not a container parse. Recovered documents were never opened.

Some values come from small screenshots and are marked [VERIFY].

Appendix A: Commands used

A.1 Hashing (all three images, at receipt and at close)

```
certutil -hashfile <image> MD5

certutil -hashfile <image> SHA256

Get-FileHash <image> -Algorithm MD5

Get-FileHash <image> -Algorithm SHA256
```

A.2 Structure and identity

```
mmls <image>

mmls Windows_Evidence_002.dd

fsstat <image>

fsstat -o 2048 Windows_Evidence_002.dd

fsstat -o 1026048 Windows_Evidence_002.dd

img_stat <image>
```

A.3 Following a file through layers (EV-001 /etc/crontab)

```
ifind -n /etc/crontab Linux_Evidence_001.img     -> inode 127733

istat Linux_Evidence_001.img 127733

icat  Linux_Evidence_001.img 127733 > crontab_icat.txt

certutil -hashfile crontab_icat.txt MD5

istat Linux_Evidence_001.img 46066               (HideMarkedCols.xls)

icat  Linux_Evidence_001.img 46066 > HideMarkedCols_icat.xls
```

A.4 NTFS system files and deleted data (EV-002)

```
istat Windows_Evidence_001.dd 0 ... 9            (MFT entries 0 to 9)

fls -r -p Windows_Evidence_001.dd  > fls_all.txt

fls -r -p -d Windows_Evidence_001.dd > fls_deleted.txt

ils -r Windows_Evidence_001.dd

ffind Windows_Evidence_001.dd 96

istat Windows_Evidence_001.dd 96

icat  Windows_Evidence_001.dd 96 > inode96_icat.docx

blkstat Windows_Evidence_001.dd 346899
```

```
tsk_recover Windows_Evidence_001.dd <Exports\Recovered_EV002>
```

A.5 Block checks (EV-001)

```
blkstat Linux_Evidence_001.img 776861

blkstat Linux_Evidence_001.img 64957

ifind -d 776861 Linux_Evidence_001.img           (failed: Invalid walk range)
```

A.6 Timelines

```
fls -r -m "/" -f ntfs Windows_Evidence_001.dd > body_EV002_fls.txt

ils -r -m Windows_Evidence_001.dd > body_EV002_ils.txt

mactime -b body_EV002_fls.txt -y -z UTC > timeline_EV002_UTC.txt

mactime -b body_EV002_fls.txt -y -z UTC -d > timeline_EV002_UTC.csv

mactime -b body_EV002_fls.txt -y -z UTC -i day <index> > ...

mactime -b body_EV002_fls.txt -y -z UTC 2019-12-19..2019-12-20

fls -r -m "/" Linux_Evidence_001.img > body_EV001_fls.txt 2> fls_err.txt

mactime -b body_EV001_fls.txt -y -z UTC -d > timeline_EV001_UTC.csv
```

A.7 Carving (GUI tools)

```
WinHex: Tools > Disk Tools > File Recovery by Type (read-only mode;

        OLE2/MS Office, Office 2007+, RTF, PDF; subfolder per type)

PhotoRec: source = Linux_Evidence_001.img; Paranoid; Keep corrupted = Yes;

          free-space run and whole-disk run
```

A.8 PowerShell checks

```
Get-ChildItem ... | first 8 bytes, size, MD5 -> carved_inventory.csv

Test-Signature (signature check of the 22 files) -> sigcheck_EV002.csv

(string search for WordDocument / EncryptionInfo / EncryptedPackage)

find /c /v "" <file>        findstr / Select-String
```

Conclusion

EV-001 is a single Ext4 volume from an Ubuntu 16.04 LTS virtual machine, roger-Virtual-Machine, installed on 9 June 2016, set to Asia/Kolkata and stopped without a clean unmount later that day. Its scheduled task file holds only defaults. HideMarkedCols.xls was introduced onto the system with its 2011 modified date preserved. Its hash, raw location, carved copies, slack and timeline entries all agree. The EV-001 image is incomplete, so other users' data cannot be assessed and must not be treated as deleted.

EV-002 is a 2 GiB NTFS volume with no partition table, and EV-003 is an MBR disk with two NTFS partitions at sectors 2,048 and 1,026,048. On EV-002, 22 deleted files were enumerated and recovered by metadata. Fourteen match their signatures, three are OLE2 containers carrying encrypted-package markers, and five are plain text. In every recovered entry, the MFT-modified time precedes the creation time, a systematic pattern whose cause is undetermined.

Every image matched its baseline SHA-256 at every check, so the evidence is shown to be unchanged. The open items in Sections 12 and 13 are carried into later modules.

